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THE COST OF CAPITAL

7

INTEREST RATES ON THE MOVE

In May 2023, the South African Reserve Bank raised its short-term interest rate (the REPO rate) to 8.25%, which according to Reuters, is its highest level in 14 years. From November 2021 to May 2023, the REPO rate has risen from 3.5% to 8.25%. Long-term interest rates will often be higher than short-term interest rates. The rise in interest rates, particularly long-term risk-free interest rates, such as government bond yields, will lead to the increase in a firm's cost of capital as the government bond yield is the base rate employed to determine a firm's cost of equity and its cost of debt. Banks will add a risk premium to either the REPO rate or the ZARONIA (previously JIBAR) rate in order to determine the cost of short-term debt, and banks will add a risk premium to the government bond yield to determine a firm's cost of long-term debt. The increase in interest rates has significantly raised the weighted average cost of capital(WACC) of South African companies. As this is the discount rate employed to determine the present value of a project's cash flows, a significant increase in the WACC will reduce the value of projects and may lead to companies cutting back on investments. Higher interest rates will also put pressure on consumers and a firm's sales may also be affected by higher interest rates. It is not only the risk-free interest rate that affects a firm's WACC. It is also the risk of the industry sector and specific firm risks relating to a firm's level of financial leverage, volatility of the firm's returns relative to the market, downside risks, firm size, and other variables that affect afirm's WACC. We will explore all these factors in this chapter.

LEARNING OBJECTIVES

After working through this chapter, you should be able to:

- Understand the concept of the weighted-average cost of capital (WACC).
- Determine the cost of debt.

- Determine the cost of preference share capital.
- Calculate the cost of equity using the dividend growth model and the CAPM approach.
- Understand the practical issues of estimating the CAPM parameters.
- Understand how a firm's capital structure affects a firm's WACC.
- Calculate a firm's WACC.
- Understand the use of the WACC in determining a firm's Economic Value Added (EVA).
- Understand alternative methods of determining the cost of equity.
- Determine the cost of capital of divisions, including unlevering and relevering betas.
- Reflect on adjustments made in practice to the cost of equity.
- Comprehend the evidence of the market risk premium based on surveys and historical returns.
- Understand why companies use hurdle rates that differ from its WACC to evaluate projects.
- Reflect on the empirical evidence of CAPM.
- Understand how to use the Fama-French Three-Factor model to determine a company's cost of equity.

INTRODUCTION

You will recall that the financial manager is required to create wealth through effective and efficient use of financial resources. The financial manager applies these resources, in accordance with the strategic objectives of the business, by seeking investment opportunities which achieve a return at least equal to that required by investors. If the business in question is a company, and the shareholders provide all the required funds, the rate of return required by shareholders is that company's cost of capital. Page 7-2

When the assets required by the company are not financed totally by shareholders' funds, each R1 invested emanates in some proportion from various sources of funds.

The proportion in which it is decided to use various sources of finance is the *capital structure* of the company.

For example, company A, which requires R500m in financing, may issue ordinary shares for the full amount or may decide to raise R300m through the issue of ordinary shares and the remaining R200m through the issue of debt. Deciding on the capital structure of a company is not straightforward; the decision is made by management, who try to select the option most likely to succeed in creating wealth for the shareholders. This chapter does not debate the issues relating to selection of an optimal capital structure. We will do this in Chapter 14. Many companies are of the view that an appropriate target capital structure will reduce their cost of capital. For example, Distell states that an objective of management is to "maintain an optimal capital structure to reduce the cost of capital." In this chapter we will focus on determining the overall rate of return required, given an existing target capital structure. Before we can determine a company's WACC, we first have to calculate the marginal cost of debt, preference shares and ordinary equity and then determine the weight of each financing source within the company's capital structure. If a company has equity and debt only and the cost of equity is 12% while the after-tax cost of debt is 8%, and if we assume a target debt ratio of 50%, then the company's WACC will be 10% (50% × 12% + 50% × 8%).

In order to develop the procedure, two assumptions are made. Firstly, it is assumed that a target capital structure exists for each company. This means that management accepts that an ideal proportion of equity and debt exists in the financing of the company. Secondly,fornowit is assumed that the assets into which the funds will be invested fall into a similar risk category (i.e. that the company uses one overall rate of required return). The approach to be adopted when assessing projects with risks different from that of the overall company will be discussed later in the chapter.

In this chapter, we will firstly explain the importance of the weighted-average cost of capital and explain the underlying principles. Then, we will explain the pooling of funds concept and determine the component costs such as the cost of ordinary equity, preference shares and the cost of debt. Thereafter, we evaluate the use of a target capital structure and the determination of individual weightings before we analyse such issues as divisional costs of capital and explore the important practical issues involved in setting a weighted-average cost of capital in the real world.

1 THE WEIGHTED-AVERAGE COST OF CAPITAL

Why is the weighted-average cost of capital (WACC) so important? The following are some compelling reasons for calculating a company's cost of capital:

- the evaluation of capital projects;
- the valuation of companies;
- the determination of a company's EVA or economic profit;
- the use of WACC to frame pricing decisions for industries subject to regulatory review; and
- the determination of fair value for corporate reporting purposes and impairment tests.

Firstly, a correct calculation of WACC is important to enable a company to evaluate capital projects. Does the return on a project exceed the company's required return as indicated by its cost of capital? A difference of a couple of percentage points in computing a company's WACC may have a significant impact on the level of investment undertaken by companies.

Secondly, a company's WACC enables investors to undertake DCF valuations of companies. A company's cost of capital is used to discount a firm's forecasted future operating cash flows.

Thirdly, investors are making increasing use of Economic Value Added (EVA), or economic profit, to evaluate the financial performance of companies and management is Page 7-3 increasingly evaluated on the basis of EVA. The calculation of EVA requires a company to determine the firm's WACC as EVA is determined as follows:

$$\text{EVA} = \text{Operating income} - (\text{WACC} \times \text{Invested Capital})$$

Companies are increasingly comparing their Return on Invested Capital (ROIC) to their WACC in order to evaluate whether they are adding or destroying value.

Fourthly, for companies and utilities operating in industries such as water, power and gas transmission which are subject to regulatory review, the determination of a company's WACC is critical as it is employed by the regulatory authorities to determine the company's prices.

Lastly, companies may use WACC to determine the fair value of assets for corporate reporting purposes. This is particularly relevant when there is an absence of

a liquid, orderly market for a company's assets.

Cash flows from operations will flow through the firm to all the providers of finance and we need to use a discount rate that recognises and includes the required returns of both shareholders and debt holders.

2 THE WEIGHTED-AVERAGE COST OF CAPITAL – PRINCIPLES AND FORMULA

What are the principles for determining a firm's cost of capital? The following are some of the principles to follow in calculating a company's cost of capital.

Use a weighted-average of the costs of the different sources of finance. We are interested in determining the required return that will provide a return to all the providers of financing to the firm. If we use only the cost of a specific source of finance then this will lead to flawed investment decisions, as indicated later in Example 7.1.

Use marginal costs. What is relevant is the cost of new financing. Therefore, use the interest cost of raising new debt and not the average cost of past debt issues and loans. Do not use the average interest rate on bonds and loans reported in the company's financial statements. Use data that refers to the future in calculating the cost of equity and the cost of debt.

Include the effects of corporate tax. The discount rate should be stated after corporate tax. When we discount future cash flows we discount cash flows after corporate tax and therefore to be consistent, we should use a discount rate that is stated after corporate tax. In summary we use an after-tax discount rate to discount after-tax cash flows.

Use nominal rates. Future cash flows are often stated in nominal terms and therefore we should use a nominal rate as a discount rate. Of course, real cash flows can be discounted at the real required return, but often this creates further complications in

relation to differential growth rates and determining the value of depreciation deductions.

Use market values or a target capital structure. The use of market values is more relevant in determining the true cost of each financing source. For example, a company may have fixed property stated at cost on the company's Statement of Financial Position. Let's assume that Page 7-4 this is R10m and the market value of the property is R30m. When the company approaches a bank for debt financing, it is the market value of such property that is relevant in deciding on the cost and amount of the loan. The real value that is at risk for each provider of finance is the market value of the equity and debt. Ignore book values to set the capital structure unless it reflects the future capital structure of the firm or is close to market value. It is useful to use a target capital structure if management expects the future weightings of debt and equity to change.

What is the formula that we should use to determine a firm's cost of capital? The formula should reflect the after-tax cost of each source of finance weighted by its contribution to the value of the firm.

$$WACC = K_d (1 - t)(D/V) + K_e (E/V) \qquad \textit{(Formula 7.1)}$$

where:

K_d = cost of debt

K_e = cost of ordinary equity

t = marginal company tax rate

D = market value of debt

E = market value of ordinary equity

V = market value of the firm

If the company has issued preference shares this would be included as a separate component in the above formula.

3 THE POOLING OF FUNDS APPROACH

In order to determine the appropriate discount rate for evaluating projects, the concept of *pooling of funds* requires explanation. In practice it is unlikely that every individual project will be financed in proportion to the capital structure weightings. Rather,

quantities of debt and equity are likely to be used in convenient amounts which reflect the target ratio only on average over time, as shown in Figure 7.1.

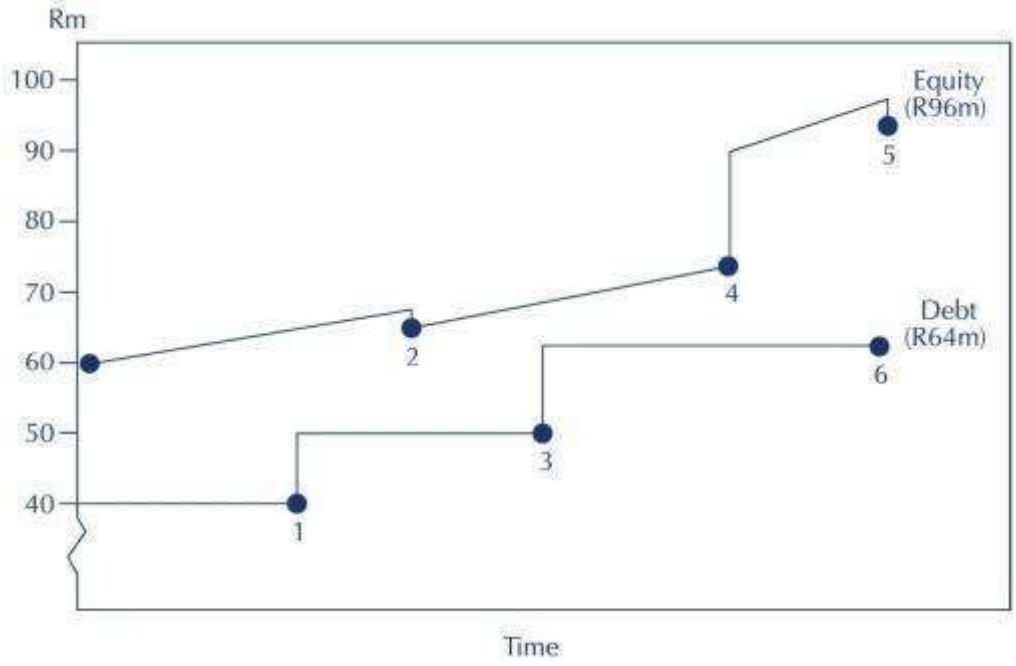

Figure 7.1 Financing of company growth

In Figure 7.1, the debt–equity ratio commenced at R40:R60m (66.67%). The equity line slopes gradually, indicating the retained income continuously generated through business operations. When further funds were required to finance new projects, only debt was issued (point one). This resulted in a move away from the target capital structure, with debt being higher than the target Page 7-5 proportion. This was slowly corrected by the generation of retained income until a dividend was paid (point two). For a while, equity funding was at a level higher than the target proportion.

In order to retain the approximate target ratio, further debt was issued (point three). The equity proportion continued to rise as a result of internally generated retained income, but a new issue of shares was necessary in order to restore the ratio (point four). In terms of stated policy, dividends are again paid (point five). Over time, however, the proportion approximates the target of 66.67% (2/3).

This step-wise procedure for raising finance may lead to the misconception that particular projects have been financed from specific funds. So, for example, if a company is at the stage of raising equity to finance a particular project, there is a danger of reasoning that the project must yield a rate of return at least equal to the cost of equity funding. Such reasoning can lead to the incorrect investment decision being made.

Let's use Sasol as an example of a real company that faced a similar situation. Sasol indicated a target gearing ratio of between 20% and 40%. The gearing ratio is defined as net borrowings (total borrowings less cash) divided by shareholders' equity. Yet in 2016, its gearing ratio was only 14.6% and had been negative in prior years due to its cash balances. Due to significant capital investments from 2015, Sasol increased its use of debt so that its gearing ratio (debt/equity) was 26.7% in 2017 and 43.2% in 2018. Sasol stated the following in its 2018 Annual Financial Statements: "The group's targeted gearing ratio is between 20% and 40%, and has been temporarily lifted to 44% until 2019 and thereafter to 40%." This reflects the lumpiness of financing decisions and explains short-term movements away from the company's target capital structure. Figure 7.2 reflects that Sasol was within its target capital structure within the three years 2006 to 2008 and within the years 2017, 2018 and 2022. The high cost of its expansion projects, particularly in the USA, and subsequent asset sales resulted in significant shifts in Sasol's debt to equity ratio from 2019 to 2023. Sasol indicated that in the long term, its gearing ratio will move to 30%.

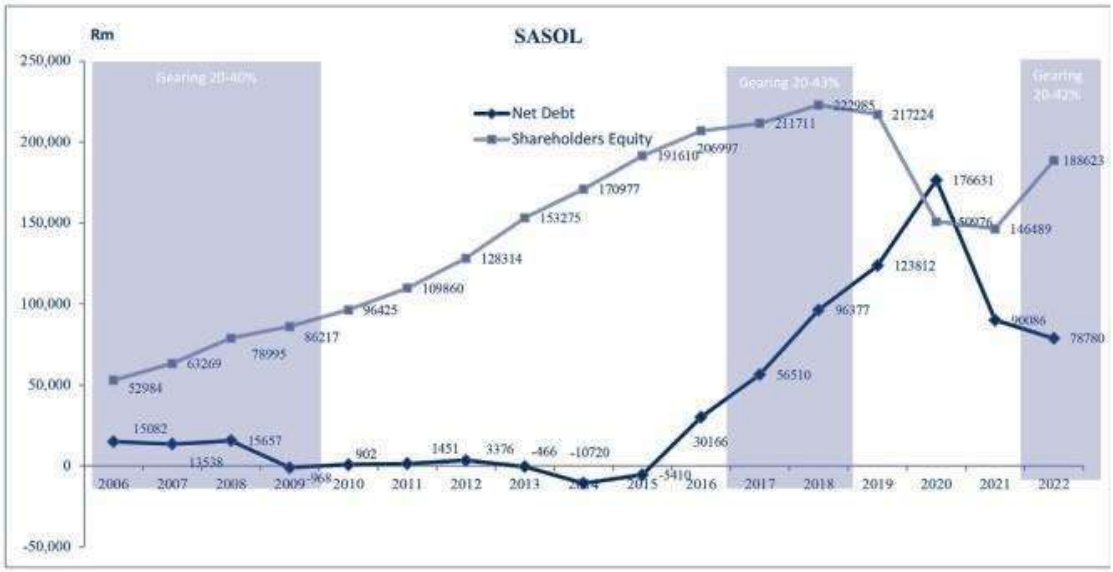

Figure 7.2: Capital structure – Sasol's net debt and equity

When evaluating projects, why is it so important not to match specific projects with the specific form of financing used to finance such projects? We will use an example to demonstrate why it is so important to use WACC to evaluate projects. Page 7-6

Example 7.1: Deciding on acceptable projects

Logic Ltd has a target capital structure of 50% equity and 50% debt. The after-tax cost (the actual cost to the company) of future debt is 8%, while the cost of future equity is

14%. The financial manager is currently considering Project A with an expected return of 10%, the funds for which will be raised by issuing debentures. Should Project A be undertaken?

If debt with a cost of 8% is used to finance Project A, which has an expected return of 10%, accepting the project appears to be a reasonable course of action. However, assume Project B comes up for consideration a month later with an expected return of 12%. In order to retain the approximate target capital structure, Project B will need to be financed by the issue of new equity at a cost of 14%. Because the cost of equity exceeds the expected return, it could be reasoned that Project B must be rejected.

The net effect is to accept the project with an expected return of 10% and to reject the project with an expected return of 12%. This is clearly an irrational set of decisions. As a target structure is retained in the long term, the cost of capital must be a weighted-average of both components of the capital structure, as determined in Table 7.1.

Table 7.1

Component	Cost	Weight	Contribution
Equity	14%	0.50	7%
Debt	8%	0.50	4%
Weighted-average cost of capital	11%		

Why does the company not issue more debt to finance Project B? The decision to issue debt to finance Project A means that the company has used up its capacity to issue further debt in the future, unless it issues more equity or uses retained earnings. Apart from reluctant bankers, this is required to ensure that the firm's capital structure falls back in line with the company's target capital structure, thereby maximising the firm's value. The capital structure decision is explained more fully in Chapter 14.

Applying a composite (weighted) cost of capital of 11% to all projects would result in Project A (with an expected return of 10%) being rejected and Project B (with an expected return of 12%) being accepted, regardless of the specific source of funds which may be used for either project. This reasoning is referred to as the *pooling of funds theory*. The principle is that a weighted cost of capital, which reflects the weighted costs of the components of the capital structure, must be used as the hurdle rate for evaluating new investment projects. It is the *marginal cost* for each component that is appropriate for the calculation – the cost of every new R1 that must be raised. Having established the fact that a weighted-average approach is appropriate for

determining the cost of capital (WACC), the following steps are required to perform the calculation:

- calculate the marginal component cost for each category of funds;
- establish the relevant weight to be used for each component; and
- determine the contribution of each component, adding them in order to obtain the weighted-average cost of capital (WACC).

These steps are outlined in the following three sections. Thereafter some additional factors are introduced.

4 COMPONENT COSTS OF CAPITAL

In Chapter 6 a number of formulae were presented for valuing financial assets. In each case it was assumed that the relevant required rate was known. In this chapter the unknown variable is the required rate. The objective is to establish methods and formulae for determining the cost of Page 7-7

each individual component in the financing mix in order to establish the rate. As a starting point we will use the financial data of Hypothec Ltd, obtained from Table 7.2 which represents its summarised Statement of Financial Position. We will assume at this stage that the rand amounts closely reflect market values.

Table 7.2 Assets and sources of finance

Statement of Financial Position	Rm
Assets	
Non-current assets	800
Current assets	300
	1,100
Equity and Liabilities	
Equity	
Share capital	250
Retained profits	350
Non-current Liabilities	
10% Debentures (3m debentures at R100)	300
12% Preference shares (issue price = R1 each)	100
Current liabilities	
Trade and other payables	100
	1,100

The top section represents the assets – i.e. the investments made by Hypothec Ltd in operating assets which must generate returns – while the bottom section represents the

sources used to finance the investments. Although there are a number of itemised sources of finance, we will only focus on the long-term permanent sources of financing, being ordinary equity, debentures (bonds) and preference shares. The preference shares are classified as a liability. In South Africa, the preference shares are often classified as debt due to the terms of issue. The debentures are redeemable in five years' time. We will now explain each source of financing:

- ***Long-term loans in the form of 10% debentures***. Debenture holders will receive a payment of R10 each year for each debenture held until the capital sum is repaid. At that stage Hypothec Ltd would need to have R300m available in cash resources – which is unlikely in a growing company – or find other sources of finance from which to fund the debenture repayment. These sources would be a new debenture issue or some other form of longterm loan. Depending on the prevailing interest rates, the applicable interest rate will be negotiated with the investors.
- ***Preference shares that pay a dividend of 12 cents per share each year in perpetuity***. There are numerous categories into which preference shares can fall. The procedures for valuation when they are redeemable have already been considered in previous chapters.
- ***Shareholders' equity*** is the sum of all items that provide the owners with a claim against the assets of the company. It consists of the original amounts invested by shareholders plus all accumulated profits which have not been paid out in the form of dividends. There are 100m shares with a total net asset value of R600m. The shares have a book value of R6 each.

The necessary formulae and methodology for determining the cost of each component are required as the first step toward establishing a weighted average cost of capital to be used as the minimum required rate of return for capital projects. Page 7-8

Cost of new debt (K_d)

Debt, for the purpose of determining the cost of capital of a company, refers to interest-bearing loans. Such loans may be made on various terms but can be divided into two categories, namely *term loans at a negotiated rate* and *debentures (bonds) at a coupon rate*.

In both cases the interest rate will be market related on the date of issue. In the case of a term loan the quoted rate will be related to the market rate currently being

charged for similar loans. In the case of debentures the rate will be established in terms of the discount or premium on par value at the time of sale. For example, if debentures of R100 have a coupon rate of 8% and market interest rates rise to 11%, the debenture will be sold at a discount price which will provide an 11% return to the lender. The impact of fluctuating market rates on debentures was discussed in Chapter 6.

Although most term loans represent variable rate financing, a company can also enter into a term loan with a fixed interest rate. Furthermore, companies are able to undertake interest rate swaps, thereby moving from a fixed interest rate to a variable rate or *vice versa*. A company can also issue notes at a variable interest rate.

From the standpoint of the company raising the finance, the before-tax cost of debt is expressed as an interest rate. The real cost to the company, however, is lower than that rate if the company does not have, or expect to have, an assessed tax loss. Because interest is normally a deductible expense for taxation purposes, the South African Revenue Service (SARS) is in effect subsidising the interest payment to the extent of the marginal tax rate of the company. The relevant component cost of debt to be used when determining the composite cost of capital is:

where:

$$K_d = I(1-t) \qquad \textit{(Formula 7.1.1)}$$

K_d = the cost of debt

I = the interest rate payable

t = the marginal tax rate

The determination of the after-tax cost of debt can be straightforward or complex depending on the terms of the financial arrangement. Assume a company borrows R200 million from a bank with repayment due in five years' time of R200 million. The bank is charging a variable rate such as prime less 1%. Let's assume that prime is 10% and that the firm is expecting to pay 9% per year. The marginal tax rate is 27%.

$$\text{After-tax interest rate} = 9\% \times (1 - 0.27) = 6.57\%$$

We will extend the calculation of the after-tax cost of debt to bonds and we will apply Section 24J of the Income Tax Act to determine the after-tax cost of debt.

Example 7.2: Calculating the component cost of debt

Hypothec Ltd wants to issue 10% debentures of R100 each. If the market-related interest rate for debentures in a similar class has risen to 11% and the company's marginal tax rate is 27%, determine the component cost of this issue. It is estimated that a total amount of R60m could be raised from an issue of these debentures.

It is clear that lenders will not be prepared to pay R100 for each debenture as the coupon of R10 will provide a return of less than 11%. The debentures will therefore be sold at a discount. The price will be determined by a valuation process which takes into account both the coupon interest and the timing of the capital repayment as discussed in Chapter 2. The price of each debenture will be in equilibrium when the company is paying a before-tax interest rate of 11%. The component cost will therefore be: Page 7-9

$$K_d = I(1 - t)$$
$$= 11\%(1 - 0.27) = 8.03\%$$

If we assume, firstly, that the debentures are non-redeemable[1], then it is possible to establish the amount which would be raised from each debenture as follows:

Value of debenture:

$$V_d = \frac{\text{coupon rate}}{\text{required rate}} \times \text{face value}$$

$$= \frac{10\%}{11\%} \times R100 = R90.91$$

Let's assume that the debentures are not perpetuities and have a redemption date in five years' time at their par value of R100 each. The current price of the 10% debentures is R93.80. What is the interest rate implicit in this price? We need to compute the yield to maturity (YTM). The YTM is equivalent to the IRR of the debenture (bond) cash flows and so we determine the IRR of the debentures (bonds) as follows:

Table 7.3 Determination of the Yield to Maturity (YTM) of the debenture (bond)

	A	B	C	D	E	F	G	H
1	Bond price	93.80						
2	Face Value	100						
3	Coupon rate	10%						
4	Year		20.0	20.1	20.2	20.3	20.4	20.5
5	Bond price		93.800					
6	Coupon payment			-10.000	-10.000	-10.000	-10.000	-10.000
7	Redemption of face value							-100.000
8			93.800	-10.000	-10.000	-10.000	-10.000	-110.000
9	YTM (IRR)		11.7075%					
10	Excel formula in Cell C9		=IRR(C8:H8,0.11)		[Note: 0.11 is an estimate - a starting point for Excel]			

Using Excel, we determine that the yield to maturity (YTM) is currently 11.7075% for debentures issued by the company. Assuming immaterial issue costs, and tax based on the yield to maturity[2], then the after-tax cost of debt will be:

$$K_d = 0.117075(1 - 0.27) = 8.55\%$$

[Using a financial calculator: N = 5, PV = 93.80, PMT = –10, FV = –100, $\therefore i = 11.7075\%$]

The cost of debt and Section 24J of the Income Tax Act

We multiplied the bond's YTM by (1–tax rate) to determine the after-tax rate. We can also determine the tax cash flows and determine the after-tax yield to maturity. Section 24J of the Income Tax Act sets out the calculation of the accrual of interest from a lender's perspective as well as setting out the incurrence of interest from a borrower's perspective. Interest is spread out over the period of the transaction on a yield to maturity basis. A company is required to use the yield to maturity to determine the incurrence or accrual from a tax perspective. Any deduction is determined by multiplying the bond's yield to maturity by the opening balance Page 7-10 rather than using the cash flow for a specific period to reflect the interest tax deduction for that period. The formula used in Section 24J is stated as follows:

$$A = B \times C$$

A = the amount to be determined

B = yield to maturity

C = adjusted initial amount

The initial amount is defined as the issue price or the transfer price. The adjusted initial amount is equal to the initial amount plus accrued interest to date less payments made to date. Let's go back to our example and use the calculations required by Section 24J to determine the financing cost deduction for each year. We computed the YTM to be 11.71%; actually we will use a less rounded YTM, being 11.7075% and

apply this to determine the yearly tax deduction arising from the issue of debentures (bonds).

Table 7.4 Determination of Section 24J interest deduction

	B	C	D	E	F	G
46		a	b = a x YTM	c	d = a+b-c	b x tax rate
47	Year	Open. bal	S24J	Coupon	Cl. bal	Tax saving
48	20.1	93.8000	10.9816	-10.0000	94.7816	2.9650
49	20.2	94.7816	11.0966	-10.0000	95.8782	2.9961
50	20.3	95.8782	11.2249	-10.0000	97.1031	3.0307
51	20.4	97.1031	11.3683	-10.0000	98.4715	3.0695
52	20.5	98.4715	11.5285	-10.0000	100.0000	3.1127
53			[Ex. 93.8000 x 0.117075 = 10.9816]			

The opening balance in 20.1 reflects the proceeds per bond that the company will receive on the issue or sale of the bonds. For years from 20.2, the opening balance represents the adjusted initial amount of the bond. The Section 24J deduction for year 20.3 for example, is R11.2249 (95.8782 X 0.117075), yet the company only makes a cash flow payment of R10.00 in interest for the year. The tax saving of the S24J interest deduction is determined by multiplying the deduction by the marginal tax rate of 27%. We can now determine the after-tax YTM at 8.546% (rounded) as set out in Table 7.5.

Table 7.5 Determination of the after-tax yield to maturity (YTM)

	A	B	C	D	E	F	G	H
37			20.0	20.1	20.2	20.3	20.4	20.5
38	Cash flows		93.800	-10.000	-10.000	-10.000	-10.000	-10.000
39	Redemption of face value							-100.000
40	Tax flows (deduction x tax rate)			2.9650	2.9961	3.0307	3.0695	3.1127
41	After-tax cash flows		93.8000	-7.0350	-7.0039	-6.9693	-6.9305	-106.8873
42	YTM		8.546%					

If the company pays a raising fee to the bank, this may be deductible in the year incurred if the funds are to be used to finance working capital. We would include this fee, let's assume R0.938m, as an outflow in 20.0 and the tax credit of R0.25326m (27%) as an inflow probably one year later in 20.1. This would increase the YTM in Table 7.5 to 8.734%. If the raising fee is not tax deductible, because the loan will be used to acquire capital assets, then the YTM and the cost to the company will rise slightly to 8.797%. Any premium payable on redemption would be included in the Section 24J pre-tax yield to maturity calculation in order to determine the yearly interest deduction.

What happens if the company issues zero coupon bonds? In this case there are no interest payments over the term of the bond, yet interest is deductible in terms of

Section 24J. For example, assume a company issues zero coupon bonds priced at R52 each which will be redeemed at R100 in five years' time.

Table 7.6 The before-tax yield to maturity (YTM) of a zero coupon bond

	A	B	C	D	E	F	G	H
69	Year		20.0	20.1	20.2	20.3	20.4	20.5
70	Bond price		52.000					
71	Coupon payment			0.000	0.000	0.000	0.000	0.000
72	Redemption(face value)							-100.000
73			52.000	0.000	0.000	0.000	0.000	-100.000
74		YTM (IRR)	13.9723%					

In terms of Section 24J, the company will be able to make interest deductions even though the firm will not be making any interest payments over the next five years. This is set out in Table 7.7 (column D).

Table 7.7 Determination of Section 24J interest deduction

	B	C	D	E	F	G
79		a	b = a x YTM	c	d = a+b-c	b x tax rate
80	Year	Open. bal	S24J	Coupon	Cl. bal	Tax saving
81	20.1	52.000	7.266	-	59.266	1.962
82	20.2	59.266	8.281	-	67.546	2.236
83	20.3	67.546	9.438	-	76.984	2.548
84	20.4	76.984	10.756	-	87.741	2.904
85	20.5	87.741	12.259	-	100.000	3.310

Table 7.8 The after-tax yield to maturity (YTM) of a zero coupon bond

	A	B	C	D	E	F	G	H
88			20.0	20.1	20.2	20.3	20.4	20.5
89	Cash flows		52.000	-	-	-	-	-
90	Redemption of face value							-100.000
91	Tax flows (deduction x tax rate)			1.962	2.236	2.548	2.904	3.310
92	After-tax cash flows		52.000	1.962	2.236	2.548	2.904	-96.690
93		YTM	10.200%					

If we look at this from the lender's perspective, then the lender will be paying tax on interest that has not been received. Therefore, Section 24J has cash flow consequences for the lender as tax is payable on interest that will not be received until year 20.5.

Cost of preference shares (K_p)

The agreement with preference shareholders provides that they will receive a dividend payment before any payment is made to the ordinary shareholders. Although the dividend payment to preference shareholders is not a fixed obligation in the same sense as interest payments on debentures, the expectation is that dividends will be paid regularly. The expectation is based on the knowledge that a company failing to pay preference share dividends may not legally pay ordinary share dividends. Moreover,

voting rights must be granted to preference shareholders while their dividends are in arrears. In addition, failure to pay preference dividends would seriously damage the creditworthiness of the company, thus reducing the opportunity for raising further debt. The same principle of market-related rates applies to preference shares as to debentures and loans.

Example 7.3: Calculating the component cost of non-redeemable preference shares

Hypothec Ltd wishes to issue 12% preference shares at an issue price of R1 each. If the current market rate for preference shares in the same class has risen to 14%, determine the component cost of this issue and the number of preference shares to be issued. The company has a marginal tax rate of 27% and will be required to pay flotation costs of 5 cents per share issued. It is known that a maximum amount of R25m could be raised from this issue.

The preference share issue has many similarities to that of the debenture issue. As market rates are considerably higher than the preference share dividend, investors will be prepared to pay an amount which returns 14% on their investment, calculated as follows:

$$\text{Value of preference share} = V_p = \frac{\text{expected dividend rate}}{\text{required rate}} \times \text{issue price}$$

$$= \frac{12}{14} \times R1 = 85.71 \text{ cents}$$

As the flotation cost is 5 cents per share issued, the flotation rate, F, is calculated as follows:

$$F = \frac{5}{85.71} = 5.83\%$$

While the net receipt is 85.71 - 5.00 = 80.71 cents per share, in order to derive a general formula, this can be expressed as:

$$\text{Net receipt} = V_p(1 - F)$$

$$= 85.71\ (1 - 0.0583) = 80.71$$

A formula for determining the component cost of non-redeemable preference shares is:

$$K_p = \frac{D_p}{V_p(1-F)} \qquad \textit{(Formula 7.2)}$$

$$= \frac{12}{85.71\ (1-0.0583)} = 14.87\%$$

In order to determine the cost of redeemable preference shares, we would undertake the same calculations as the determination of Yield to Maturity for debentures (bonds).

It will be noted that no tax shield is provided by the payment of preference share dividends. The marginal tax rate is therefore not a relevant variable when determining the component cost of preference shares. However, South African companies have been able to raise capital by issuing preference shares to the banks at a variable rate. As preference dividends are not taxable for companies, South African banks have been ready to invest in preference shares issued at a variable dividend rate of about 70%–75% of the prime rate. Although the marginal tax rate is not directly relevant, it is considered by the banks in setting the required dividend rate. Page 7-13

Cost of shareholders' equity

The cost of funds contributed by shareholders is generally more difficult to ascertain. The cost of equity is the rate of return required to at least maintain the value of the shares. If the facts to be outlined in Example 7.4 are known about Hypothec Ltd, a number of alternative methods may be used in order to establish the required rate of return on shareholders' funds currently invested – which is referred to as the *cost of retained earnings* (k_r). Although the cost of equity is indicated as k_e in our general formula, we have further divided this to reflect the cost of new shares, k_s, and the cost of using retained earnings, k_r.

Example 7.4: Calculating the component cost of retained income

The following facts are known about Hypothec Ltd:

- Current market price: R7.68
- Expected EPS for the next year: R1.28
- Dividend payout ratio: 30%
- Expected growth rate: 11% p.a.
- Beta: 1.2

In addition, it is established that the expected return on the market (R_m) equals 14% and that the risk-free rate (R_f) equals 8%. This means that the market risk premium is expected to be 6%.

The component cost of equity to be used in the WACC calculation for Hypothec Ltd can be determined using a number of different methods, all of which require some estimations. Two of the most commonly used methods will be illustrated, both of which could be used before a cost is determined. You are referred to Chapter 6 on valuations for a more complete treatment of ordinary share valuation.

Dividend yield and growth method

In this case, the well-known dividend growth model is appropriate. In order to apply this model, the expected dividend (D_1) must be determined. As the expected EPS is R1.28, and the dividend policy is known to be 30% of earnings, D_1 is estimated as follows:

$$D_1 = R1.28 \times 0.30 = 38 \text{ cents}$$

Applying the growth formula:

$$k_e = \frac{D_1}{P_0} + g \qquad \textit{(Formula 7.3)}$$

$$= \frac{0.38}{7.68} + 11\% = 15.95\%$$

The assumption of constant growth is made when using this model. The factors that underpin this model, such as the determination of the growth rate, are more fully explained in the chapter on valuations, Chapter 6. If new equity has to be raised, the component cost will increase as a result of flotation charges. If it is assumed, for example, that flotation costs equal 5% of funds raised, and if the growth method is used, the appropriate formula would be:

$$k_s = \frac{D_1}{P(1 - F)} + g \qquad \textit{(Formula 7.4)}$$

where: k_s = the cost of new shares

In this example k_j is calculated as follows:

$$k_s = \frac{0.38}{7.30} + 11\% = 16.21\%$$

Capital asset pricing model method (CAPM)

In this case it is necessary to accept the assumptions that underlie CAPM, particularly the assumption that investors hold diversified portfolios. The CAPM approach states that the company's cost of equity is dependent upon the risk-free rate plus the market premium multiplied by the company's beta. The firm's beta measures the level of systematic risk. If we analyse this more carefully, then all we are saying is that investors will demand to earn a return at least equal to the return offered by government bonds plus a risk premium. Investing in equities is risky and therefore investors require a market premium for investing in the equities markets, which will reflect volatile returns in relation to government bonds. However, a particular company may be more or less risky than the market and so the market premium is adjusted by the relative volatility of the company to the market, which is measured by the company's beta. However, CAPM assumes that the only risk that is relevant is systematic risk. This is indicated by a company's beta, which demonstrates the volatility of a share's returns relative to the market. A beta of 1.20 indicates that Hypothec is more risky than the market. This variability cannot be diversified away. The market risk premium is indicated by ($R_m - R_f$). Using the information available, the necessary calculation is:

$$k_e = R_f + \beta(R_m - R_f) \qquad \textit{(Formula 7.5)}$$

where: k_e = the return required by equity holders

R_f = the risk-free rate

b = the beta of the share

R_m = the return on the market portfolio

In this example, k_r is calulated as follows:

$$\begin{aligned} k_e &= 0.08 + 1.20\ (0.06) \\ &= 0.08 + 0.072 \\ &= 15.2\% \end{aligned}$$

In the above calculation 8% represents the risk-free rate and 6% a premium for market risk attached to shares in Hypothec Ltd. Where information is not available to apply the CAPM, the same rationale could be used, but the 7.2% would have to be determined on a more subjective basis. In South Africa, the PwC Valuation Methodology Study 10e (December 2022) indicated that the capital asset pricing

model (CAPM) was the most widely used model to determine the cost of equity. The survey found that 79% of respondents always used CAPM and about 18% frequently used CAPM. Therefore, companies make dominant use of the CAPM to determine a company's cost of equity. Of course this then leads to the issue of how best to estimate the riskfree rate, the market premium and the company's beta. We will explain some of the practical issues and limitations of estimating the CAPM parameters later in the chapter.

Bond yield plus a risk premium method

Although widely used, the CAPM approach is not always accepted by management and analysts, mainly due to the underlying assumptions of the model as well due to difficulties in obtaining an accurate beta. Some companies and analysts prefer to determine the cost of equity by using the company's bond yield or interest rate and to add a risk premium which is based

on management's judgement or an analyst's past experience. The risk premium will mostly be in the range of 2% to 6%. Remember that we are not using a risk-free rate. We are using the company's bond yield or the bond yield of a comparable company or the company's borrowing cost. The bank and investors would have included company specific risk in determining the company's bond yield or borrowing rate. For example, assume that a AAA-rated firm's bond yield is 9% and the firm adds an equity risk premium of 5%, then the cost of equity will be:

$$\text{Cost of Equity} = \text{Bond yield} + \text{risk premium} = 9\% + 5\% = 14\%$$

The cost of debt for a BBB-rated company may be set at 11% and so the cost of equity in this case would be:

$$\text{Cost of Equity} = \text{Bond yield} + \text{risk premium} = 11\% + 5\% = 16\%$$

Bond yields and interest rates of firms are set by the market and are external to the firm. That gives us confidence. Whilst the risk premium is subjective and depends on management's judgement, perhaps we should trust their experience to get us to a reasonably accurate cost of capital. After all, we rely on their judgement and experience to determine future project cash flows.

5 WEIGHTING COMPONENTS OF CAPITAL STRUCTURE

In order to calculate a company's WACC, we need to determine the weights to allocate to each cost of finance. In our example, in Table 7.1, we used 50% equity and 50% debt (Debt/Equity = 0.5/0.5 =100%). How do we decide on the weights of debt and equity to calculate WACC? This represents a company's capital structure and we will study this topic in greater depth in Chapter 14. We assume that there is an optimal or target capital structure that will maximise the value of a company and reduce its cost of capital.

The basic concepts underlying a target capital structure rest on the principle that the use of debt, which is less costly than equity, increases the financial leverage of a company. Furthermore, debt is less costly than equity and also results in an interest tax shield. The company is normally able to deduct the interest expense from taxable income. Therefore, if a company borrows R100m at 10% per year, then this will result in a tax deduction of R10m per year and a tax saving (tax shield) of R2.7m per year (R10m x 27%) assuming a corporate tax rate of 27%. If we use equity financing of R100m, then this will not result in any tax deduction. Therefore, the use of debt should increase the value of a company. However, as a company increases its debt financing relative to equity, this will increase the risks that a company faces. This relates particularly to the risk of financial distress that a company will face if the company cannot meet its interest payments. Also, higher interest rates may result in losses and then the interest tax shield is not worth very much. This happened to the Edcon group (Edgars, Red Square, Boardmans, CNA, Jet) from 2008 to 2017 as Edcon employed significant levels of debt in its capital structure[3].

Also, as the company takes on more debt, there is a greater risk that the company will not meet its loan covenants which may result in all debt becoming due and payable. As a company's debt/equity ratio increases, it will find that the cost of debt becomes increasingly expensive. So debt is good up to a point. After this optimal debt/equity ratio, the present value of all the financial distress costs will exceed the benefit of debt, leading thereafter to a fall in the value of the company.

The target capital structure is the debt/equity ratio, which leads to an optimal value of the company. The impact of increasing the debt/equity ratio on company value is depicted in Figure 7.3.

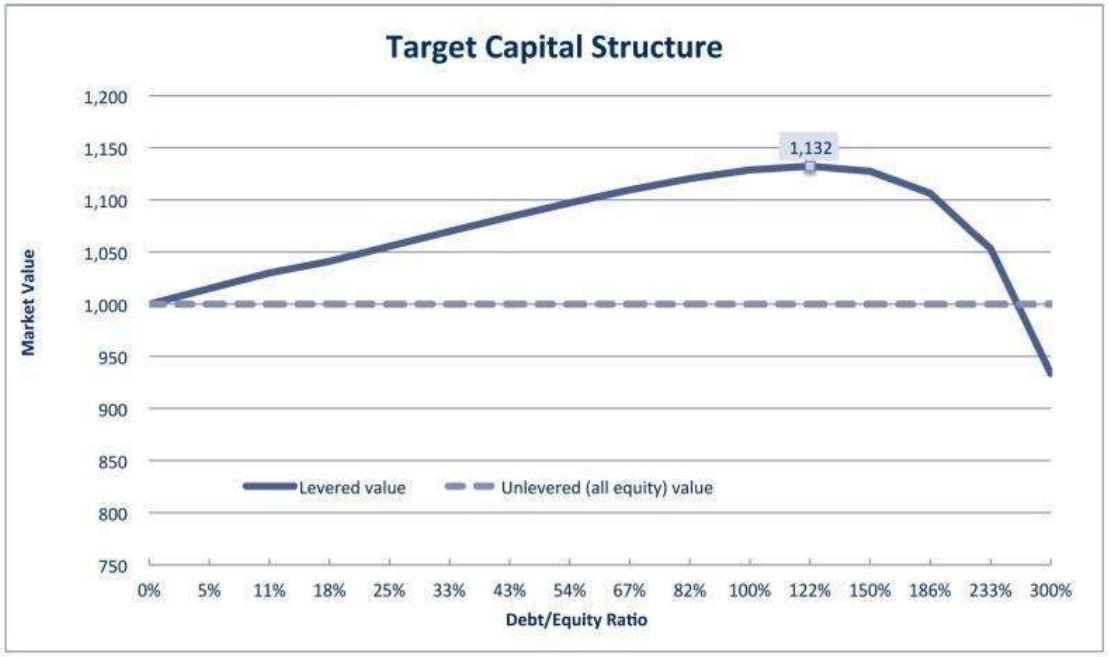

Figure 7.3 Target capital structure

The target capital structure is indicated by a debt/equity ratio of 122%, which is represented by 45% equity financing and 55% debt financing (0.55/0.45 = 122%)[4]. After this debt/equity ratio, the increase in value of the tax shields of additional debt is more than offset by the increase in the present value of the costs of financial distress.

There are three possible capital structure weights that can be used to determine a company's WACC: the **target** capital structure, the current **market values** of debt and equity financing and the accounting or **book values** of debt and equity that are disclosed in the Statement of Financial Position. The ideal approach is to first use the target capital structure, if available, or otherwise to use market values. Only use book values if this reflects the target capital structure. For example, Sasol reports that its target capital structure is a net debt/equity (gearing) ratio of 20%-40% but the group indicated that it wished to move to a net debt/equity ratio of 30%. In prior years, Sasol's actual net debt/equity ratio was less than zero, but we would use its target capital structure to determine its WACC as the group is intending to move to this debt/equity ratio over time.

If there is no target capital structure (or the target structure indicated by management is not credible), then it is best to use capital structure weights based on **market values**. Why? Assume PropCo Ltd owns a building, stated at its cost of R20m in its Statement of Financial Position, but the current market value is R80m. The company currently has no debt. Its borrowing capacity and the cost of debt will be determined by the market value of R80m, and not by the cost of R20m. Secondly, the

company should earn an appropriate return on the current market value of the property and not its cost. Furthermore, equity holders will evaluate the risk of any investment in the company by the market value of the company's assets and liabilities. Therefore, use market values rather than book values. In order to determine the market value of equity, multiply the share price by the number of ordinary shares outstanding at year end. If there is evidence that a different structure is proposed for the future, the new structure would provide the relevant weights. Page 7-17

Due regard must be given to the existence of such items as leasing contracts and permanent bank overdrafts, both of which serve to displace the opportunity for raising further debt. As a result, they must be considered to be a part of the capital structure. We will come back to leasing later in the chapter. The item "deferred tax", present on many Statements of Financial Position of companies, is a more contentious item. If it is significant, it should be included with retained income since investors are likely to expect a return on these funds. It has been stated before that market values should be used to determine the capital structure weightings. However, some companies have employed book values to determine the capital structure weightings.

Example 7.5: Finding the weights to determine the WACC

If the values of Hypothec Ltd's sources of finance reflect the target proportions, we can establish the weights to be used for the calculation of the WACC as follows:

Source	Rm	Weightings
Equity	600	60.0%
Preference shares	100	10.0%
Debt	300	30.0%
	1,000	100.0%

We did indicate that we should use market values to determine the weightings used to determine the WACC. The market values are determined as follows:

	No. (m)	Price	Market Value	Weightings
Equity	100	7.68	768.00	67.66%
Preference shares	100	0.8571	85.71	7.55%
Debentures	3	93.80	281.40	24.79%
Total market value			1135.11	100.00%

6 CALCULATING THE WACC

In the examples used so far in this chapter, we have developed the component marginal cost for the three categories of financing used by Hypothec Ltd and have

established the appropriate weightings to be used for the calculations of the WACC. These variables are now brought together in Table 7.9 in order to arrive at the WACC of 13.5%. This is the appropriate rate to be used as a basis for accepting or rejecting investment projects which are to be evaluated. Hypothec Ltd must earn a return of 13.5% on its investments. Failure to achieve this return will result in the loss of value.

Table 7.9 Calculating the WACC of Hypothec Ltd

		Book values		Market values	
Source	Cost	Weightings	WACC	Weightings	WACC
Equity	15.20%	60.00%	9.12%	67.66%	10.28%
Preference shares	14.87%	10.00%	1.49%	7.55%	1.12%
Debt	8.546%	30.00%	2.56%	24.79%	2.12%
Weighted average cost of capital		100.00%	13.17%	100.00%	13.52%
Weighted average cost of capital (rounded)			13.20%		13.50%

The use of book values would result in a lower WACC of 13.2%. We would only use book values to determine the weights if this also reflected the company's target capital structure. Page 7-18

If we use market value weights, then the WACC is 13.5%. In most cases, this is what we would do in practice unless the company has a target capital structure that it has communicated to the market, like Sasol has done. Many South African companies have capital structures that reflect low debt–equity ratios and it is probably more correct to employ a target capital structure which includes a greater use of debt in the future.

7 BREAKS IN THE WACC

As long as a company retains its target capital structure and if the costs of equity and debt remain constant, then the company's WACC will remain constant. What happens when the company needs to raise additional debt and equity financing at a higher cost? This is particularly true when a company does not have sufficient retained earnings and in order to stay within its target capital structure, it is required to issue additional shares to raise the required equity financing. The issue of new shares involves issue or flotation costs and this will increase the cost of equity. If the dividend growth model is being used to determine the revised cost of equity, then this is simple to include in the model $[D_1/P + g]$ as the price is reduced by the issue or flotation costs. If using CAPM, then make a separate adjustment to increase the cost of equity (due to the CAPM assumptions about zero transaction costs).

We will assume that in our example, the cost of equity will increase from 15.2% to 15.5% for the issue of new shares. When do we need to issue new shares and incur a higher cost? It depends on the retained earnings and the weighting of equity in the company's capital structure. In our Hypothec example, the EPS was R1.28 and the expected dividend was R0.38 per share (R1.28 × 30%). This means, we expect to generate R0.90 per share (R1.28-0.38) in retained earnings and total retained earnings that we can use to finance projects will be: R0.90 x 100m shares = R90m

The question, however, is what is the total rand amount of projects we can finance before the cost of capital will increase?

$$\text{Break point} = \frac{\text{component funds available}}{\text{weight in capital structure}} \qquad \textit{(Formula 7.6)}$$

The break point for equity is $\frac{\text{R90m}}{0.6766} = \text{R133m.}$

This means that we can invest up to a total of R133m before the company needs to issue new equity at a higher cost of equity. This is in order to retain the company's optimal capital structure, which is based on market values. What about increases in the cost of other sources of finance? Sometimes the bank may charge a higher interest rate above a certain level of borrowing. In the case of Hypothec, the bank has indicated that it will increase the interest rate to 12.8% if the incremental borrowing for the year increases above R60m. In Table 7.9, the weighting of debt is 24.79% of the company's capital structure.

The break point for debt is R242m (R60m/0.2479). We can invest up to R242m without incurring a higher interest rate. Therefore, any investment above R242m, which will be 24.79% financed by borrowings, will be at an interest rate of 12.8%, which is 9.344% after tax [0.128 × (1 – 0.27)]. We will assume that there are no limits on the issue of preference shares at 14.87%.

Example 7.6: Breaks in WACC and investments

In order to simplify our calculations, assume that the company can invest in Project A at a cost of R133m, Project B at a cost of R109m, Project C at a cost of R48m and Project D at a cost of R30m. The return per project, as indicated by each project's internal rate of return, is indicated as follows: Page 7-19

Projects	A	B	C	D
Internal rate of return	23%	21%	17%	10%

Projects	A	B	C	D
Funds required (Rm)	133	109	48	30
Cumulative funds required (Rm)	133	242	290	320

We know that Project D should not be accepted, as its return of 10% is lower than the company's lowest estimate of WACC. The other projects will be accepted but the WACC may vary slightly. The WACC of the first R133m investment will be 13.5%, the WACC used to discount the cash flows of Project B will be 13.7% as only the cost of equity has increased and for Project C, the cost of capital will be 13.9%, as determined as follows:

		Market values	
Source	Cost	Weightings	WACC
Equity	15.50%	67.66%	10.49%
Preference shares	14.87%	7.55%	1.12%
Debt	9.344%	24.79%	2.32%
Weighted average cost of capital		100.00%	13.93%
Weighted average cost of capital (rounded)			13.90%

You may argue that as the company makes larger borrowings, then the cost of financing should fall as long as the debt/equity ratio remains constant. This may be true, at least for a while. A larger company may be able to borrow from offshore sources or list overseas, thereby increasing its access to capital. For example, by listing on the London Stock Exchange, a company should be able to issue shares at a lower cost and will be able to borrow at a lower interest rate. However, as a company invests increasing amounts, then it should also be true that a company will first select the projects with higher returns and so the effective security for the bank may fall as the company expands and borrows greater amounts. Often rights issues by companies can be seen as a negative sign of the company's prospects and issuing shares is definitely more costly than using retained earnings. In practice, as we use WACC to discount future uncertain cash flows, minor adjustments to WACC will often be ignored, and it is therefore more relevant to accurately determine the CAPM parameters such as the company's beta and the market risk premium, as well as make other adjustments to determine a company's cost of equity. We will come back to this later in the chapter.

8 FUNDS FROM NON-CASH FLOW ITEMS

EPS is calculated from the accounting income after such items as depreciation and deferred taxation have been deducted. These expenses do not, however, constitute a

flow of funds out of the company. What happens to the actual funds represented by such items as depreciation? The assumption is that they are used to replace existing assets. There is a cost attached to the use of these funds as they represent real assets belonging to shareholders. The relevant cost is the cost of capital in operation before considering new capital investments.

Throughout the chapter we have referred to new capital investments, assuming that depreciation-generated funds are used for replacement. However, if the new projects include replacement of existing assets, the break in WACC can be extended by the amount of additional funds available from non-cash flow expenses that will not be reflected in the expected retained income. This additional factor does not affect the calculation of the WACC, but does delay the point at which new funds need to be sought.

9 ESTIMATING THE COST OF CAPITAL OF DIVISIONS

Until now we have determined the cost of capital of the firm. Yet a company may be operating in a number of sectors that will have varying levels of systematic risk. Therefore, a company should not use an overall company cost of capital to evaluate the investments Page 7-20

in all divisions but should rather use a divisional cost of capital. How does the company determine a divisional cost of capital? Firstly, we will need to find comparable companies that operate in the same sector as the proposed project. Although it is not always easy to find a "pure play" company, i.e. a company that operates only in a single sector; we need to find a company that comes closest in reflecting the proposed project. We then need to make adjustments for varying levels of financial leverage as the equity beta of the comparable company will include an adjustment for the comparable company's level of financial leverage. We need to "unlever" the comparable company's beta to get to the company's asset beta. We then need to "relever" the beta to reflect our company's debt–equity ratio. We will explain this in greater detail in Chapter 10 and here we only include the required adjustments to determine a divisional cost of capital.

Use the Hamada formula for unlevering and relevering betas to determine the beta to use.

$$\beta_U = \beta_L/[1 + (1 - T)(D/E)] \qquad \textit{(Formula 7.7)}$$

where: β_U = unlevered beta (asset beta)

β_L = levered beta (equity beta)

D/E = debt–equity ratio

T = corporate tax rate

Example 7.7

Q Ltd is investing in a project in a different sector which has a comparable company, Z Ltd. The debt–equity ratio of Q Ltd is 100% whilst the debt–equity ratio of Z Ltd is 150%. The equity beta of Z Ltd is 1.30. The marginal cost of debt of Q Ltd is 11%. The market risk premium is 5% and the risk-free rate is 8%. The tax rate is 27%. What beta should Q Ltd use to determine the cost of equity? What is the cost of capital that Q Ltd should use to evaluate the project?

Step 1. Determine the unlevered beta of Z Ltd

$$\beta_U = 1.30/[1 + (1 - 0.27)(150\%)] = 0.620525$$

Step 2. Relever the asset beta to reflect the financial leverage of Q Ltd

$$\beta_L = \beta_U \times [1 + (1 - T)(D/E)]$$
$$\beta_L = 0.620525 \times [1 + (1 - 0.27)(100\%)] = 1.07351$$

Step 3. Determine the cost of equity of the project

$$\text{Cost of equity} = 8\% + 1.07351(5\%) = 13.37\%$$

Step 4. Determine the cost of capital (WACC) of the project

$$\text{WACC} = 50\%[11\%(1 - 0.27)] + 50\%[13.37\%] = 10.70\%$$

We will cover unlevering and relevering betas in greater depth in Chapter 10. The principle is that if a company is expanding into a new sector, then we should use the systematic risk of the sector to determine the cost of capital and not the company's current beta to determine the project's required return. The 2022 PwC Valuation Methodology Survey found that 73% of the surveyed firms employ the Hamada formula to unlever and relever betas to take into account the effects of varying capital structures on the betas of comparable companies.

10 OPERATING LEASES, CAPITAL STRUCTURE AND THE WEIGHTED-AVERAGE COST OF CAPITAL

In terms of IFRS 16, lessee companies are required to capitalise all leases, except for shortterm leases of less than 12 months and low-value leases. The distinction between operating and finance leases for lessees is essentially no longer relevant. This has affected the Statement of Financial Position of many companies and has increased the debt–equity ratios of companies. Companies are required to determine the present value (PV) of future operating lease payments using a pre-tax cost of debt as the discount rate. This results in an increase in assets, being *right-of-use assets*, with associated liabilities, being the *present value of lease payments*. The *right-of-use* assets need to be amortised over the term of the lease. How does this affect our calculation of the WACC of a company? Let's do an example.

Example 7.8

M Ltd has a current debt–equity ratio of 50% and the company's equity is currently valued at R140m and non-current debt is R70m. The company has an operating lease which requires the company to make lease payments of R30m per year for the next four years. The current pre-tax cost of debt is 10% and the tax rate is 27%. The after-tax cost of debt is therefore 7.3%. The risk-free rate is 8%, the market risk premium is 5% and the firm's current beta is 1.2.

Firstly, we determine the current weighted-average cost of capital of the company as follows:

$$\text{Cost of Equity} = 8\% + 1.20(5\%) = 14\%$$

A debt–equity ratio of 50% means that debt will amount to 1/3 of the total of equity and debt financing of the firm and equity will be 2/3 of the total financing of the firm.

$$\text{WACC} = [(2/3 \times 14\%) + (1/3 \times 7.3\%)] = 11.77\%$$

The present value of future lease payments, using the cost of debt as the discount rate, is determined as follows:

$$\text{PV} = 30 \times \text{PVFA}\ (10\%, 4\text{yrs}) = 30 \times 3.1699 = \text{R}95.097\text{m}$$

This means that the debt–equity ratio will now increase significantly from 50% to 118% as calculated as follows:

$$\text{Debt–equity} = (95.097 + 70)/140 = 118\%$$

A debt–equity ratio of 118% is equal to a debt ratio of 54% (1.18/(1 + 1.18)). The increase in equity risk means that we need to unlever and relever the current betas as follows:

Unlevering the current beta	c **Beta**	d **Debt/Equity**	e **[1+(1-T)D/E]**	f = c/e **Unlevered Beta**
$B_U = B_L/[1+(1-T)D/E]$	1.20	50.00%	1.365	0.879
	g	h	i	j =gxi
Relevering beta to the new D/E ratio	**Unlevered Beta**	**Debt/Equity**	**[1+(1-T)D/E]**	**Relevered beta**
$B_L = B_U$ x [1+(1-T)D/E]	0.879	118.00%	1.861	1.636

Page 7-22 We can now apply the relevered beta to determine the revised WACC of the company:

$$\text{Cost of Equity} = 8\% + 1.636(5\%) = 16.18\%$$

$$\text{WACC} = [(0.4587 \times 16.18\%) + (0.5413 \times 7.3\%)] = 11.37\%$$

The capitalisation of operating leases means that there will be a fall in WACC from 11.77% to 11.37%. The lower cost of after-tax debt financing is offset to a large degree by the increase in the cost of equity due to the higher financial leverage. However, it is likely that the market already recognises the implicit leverage arising from the operating lease commitments and this may already be factored into the current beta[5]. Therefore, it is important to evaluate the impact of capitalising operating leases in relation to the betas and financing structures of comparable companies. There may be other possible complications such as the taxation of operating leases which is not on a yield to maturity basis. However, in most cases the net effect of differences in tax is not expected to have a material impact on the weighted-average cost of capital. However, it is possible that practitioners have been overstating WACC by overstating the equity weighting because the capital structure weightings have not included the present value of all leases. The capitalisation of leases corrects for this explicitly. It is also important that management communicate a credible target capital structure (based on market values) to ensure that WACC is determined accurately for a company.

11 THE WEIGHTED-AVERAGE COST OF CAPITAL – SOME PRACTICAL ISSUES

The principles underlying the cost of capital are reasonably unambiguous, yet the application is fraught with peril. The definition and estimation of the parameters of the CAPM often results in differing points of view, particularly when the theory is engineered to follow a designated financial interest. It would be unfair to leave this topic without a brief exposure to the real issues and arguments that have arisen from applying the CAPM in the real world. We want to deal briefly with a few of the issues: firstly, the estimation of the CAPM parameters and secondly, the impact of taxation on the cost of capital.

The risk-free rate

How do we measure the risk-free rate? Do we use the market yield on short-term government securities such as Treasury Bills or do we use the yield on long-term bonds? Some academics promote the use of the short-term Treasury bill rate owing to the underlying assumptions of the CAPM approach. In practice, companies tend to use the long-term bond yield on government securities. This more closely matches the term of the project being evaluated as long-term projects should be evaluated using yields on long-term government bonds. Short-term rates are significantly more volatile and the yield on, say, 10-year bonds, is more consistent with setting a long-term cost of equity. An upward sloping yield curve will mean on average a higher risk-free rate as compared to using the short-term Treasury bill rate. For example, in July 2023, the yield on 3-month Treasury bills was 8.63% as compared to the R2030 government bond which matures in 2030 and offered a yield of 10.53%.

South African firms tend to use the long-term government bond yield as the risk-free rate rather than the TB rate. The 2022 PwC Survey found that firms generally employ long-term government bond yields with a term to maturity of more than 5 years. Practitioners mostly used either the R213 (2031), the R209 (2036), or the R2032 (2032) bond yields as the risk-free rate. Respondents also use 10-year bond yields derived from the yield curve. Page 7-23

The market (equity) risk premium

Investors expect to earn a return that rewards them for taking on the higher risk involved in investing in equities as compared to bonds. The market risk premium is the additional return that equity investors expect to earn over the risk-free rate. Yet

how do we measure a forward-looking market risk premium? We can do it in two ways; we can use historical premiums based on past data and assume that this will apply into the future, or we can undertake surveys of investors and ask them what premium they expect to earn.

If we use historical returns relative to bonds, the next question is how far back do we go? What period is the relevant period? If we change the period, the market risk premium changes.

How do we measure the average market risk premium? The equity premium should be determined by analysing equity returns to long-term government bond yields to ensure consistency with the risk-free rate. Do we use a geometric or arithmetic average? The geometric return will reflect the compound return from beginning to end and is therefore dependent on the starting and ending dates. Assume no dividends and that a company's share price goes up from R10 to R15 and then falls to R12 in the next year. The arithmetic average return is (50% + (-20%))/2 = 15%. How do we compute the compound return? The ending value relative to the beginning value is 1.2 (R12/10), which reflects a compound return of 9.5% ($1.2^{1/2}$) per year. The arithmetic average is more in line with the CAPM approach but results in a higher required return than the geometric return. The geometric average is consistent with a long-term return on equities but is very sensitive to the period employed. Then there is the issue of survivorship bias. There are equity markets that failed. For example, investing in Russian equities in 1900 would have been a bad idea[6]. When we look back in time we are looking through the prism of successful companies and this may result in an overstated equity premium. Some analysts reduce the equity premium by 1-2% to take cognisance of this factor. Table 7.10 indicates the average market or equity risk premium (based on geometric means) for selected countries from 19002022 (123 years).

Table 7.10 Equity (market) risk premiums

Market risk premium 1900 - 2022	Premium vs Bonds	Premium vs Bills |
South Africa	**5.1%**	**6.0%**
Australia	5.0%	6.1%
Canada	3.7%	4.3%
France	3.4%	5.6%
Germany	5.1%	6.0%
Italy	3.2%	5.7%

Market risk premium 1900 - 2022	Premium vs Bonds	Premium vs Bills |
Japan	5.2%	6.1%
Netherlands	3.6%	4.6%
Spain	1.8%	3.3%
Switzerland	2.5%	3.8%
United Kingdom	3.8%	4.4%
USA	4.6%	5.9%
World Ex-USA	2.9%	3.8%

USA**Source: Credit Suisse Global Investment Returns Yearbook 2022 & correspondence (M. Staunton)**

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There has been a steady fall in the market risk premium over time. This may reflect changes to the tax system, such as lower tax rates, regulation of equity markets, greater information flows and lower political risks. In South Africa, the market risk premium used by analysts has tended to range from 4% to 7%. For South Africa, we can see from Table 7.10 that equities have outperformed bonds by 5.1% over 123 years. In relation to country performance, South Africa Page 7-24 is the leading country in terms of the performance of equities relative to bonds. However, it is relevant to note that for most countries, the performance of equities relative to bonds and treasury bills is much lower. The world ex-USA average (weighted by market capitalisation and GDP and using a common currency) reflects a market risk premium of only 2.9% per year.

If we compute the market risk premium from 1973 to 2022, then generally market risk premiums have fallen but this also reflects the strong performance of bonds over this period due to the fall in long-term interest rates and poor equity returns for the decade 2001-2010 for most developed countries. For South Africa, equities outperformed bonds by 4.6% per year for the period 1973 to 2022, due to a strong performance by equities, although this could also reflect the relative depreciation of the Rand and these returns are stated in terms of local currency. The market risk premium seems to be falling over time among developed countries.

A note on using historical market risk premiums

Let's evaluate the logic of using historical returns to measure required returns. If a country's equity market has shown high returns in the past in relation to bonds, then we require a higher risk premium in the future, whereas for a country that has posted low returns in relation to bonds, we require a lower premium. So we value a country

that has been good to us less than a country that has been poor to us. Where is the logic in this? What does this mean? When we look at past returns and premiums, it is really to offer us an insight about possible future returns and common sense demands that we do not allow past economic or political dimensions and boundaries, which will have impacted on historical returns, to affect the cost of equity unduly.

Surveys

As it is not possible to survey all investors, surveys are sent to brokers and portfolio managers requesting their estimate for the risk premium. The PwC survey in 2022 found that respondents use an average market risk premium of 5.2% to 6.7% in South Africa. Correia and Cramer (2008) undertook a survey of listed companies and reported an average market risk premium of 5.4% amongst 32 leading South African companies. What do we think? The use of a high market risk premium has often led to underinvestment. We consider that a market premium of 4–6% is more in line with general expectations about the future performance of equities in relation to bonds. In Australia the average risk premium reflected from such surveys was 6%. Ivo Welch has undertaken surveys in the USA of finance academics and has reported a market risk premium of 5.5%. Increasingly analysts and academics are arguing for the use of a lower risk premium.

Using the dividend growth model to determine the market risk premium

If we look at the market as one large entity then we can apply the dividend growth model to determine the cost of equity for the market and deduct the bond yield to obtain the market risk premium.

$$\text{Cost of equity} = D_1/\text{Market Value} + g$$

This is essentially the dividend yield plus the expected growth in dividends. The current dividend yield in South Africa is close to 4.5%. The next question is: how do we compute the expected growth rate? It is reasonable to argue that in the longer term, dividend growth will follow earnings growth. If we assume that the real growth rate in South Africa is expected to be 3.3% per year and the inflation rate is expected be 6%, then this means that we can expect to see a nominal growth rate of about 9.3%. If we assume that overall earnings growth will reflect the economic growth in the economy,

then the cost of equity will be 13.8% (4.5% + 9.3%). The bond yield in July 2023 was 10.5% for the 7-year government bond (R2030). The market premium

using this approach is 3.3% (13.8% - 10.5%). Whilst analysts generally prefer to use the CAPM approach, we think that the generalised dividend formula does offer us a big picture view of the whole market and should be used at least as a reasonability check to other calculations. Usually, the market premium based on this method is about 4% and the current low estimate is due to a higher bond yield, which captures we believe the higher risk ratings of emerging markets.

Other indicators of the market risk premium

There are other sources of information that can be used to estimate the market risk premium and we may need all the help we can get. We have already indicated the market risk premium reported by respondents to the 2022 PwC Valuation Methodology Survey and the results of the Correia and Cramer (2008) study. Other indicators of the market risk premium are as follows:

- KPMG in their 2011 South African survey reported an average market risk premium of 5.7%. In the 2012 KPMG survey, the MRP had increased slightly to 6.0%.
- The consultancy firm McKinsey Inc. employs a market risk premium of 5%.
- NERSA set a market risk premium of 5.3% for the revenue/tariff application by Eskom for the period 1 April 2013 to 31 March 2018.
- Merrill Lynch's survey of global institutional investors reported a risk premium of 4.6% in February 2014.
- Fernandez (2011) reviewed 150 textbooks and found that the last five-year rolling average market risk premium was 5.7%.
- In a survey of professors and practitioners in 71 countries in 2016, Fernandez reported that the median market risk premium for South Africa was 6.0% and the average market risk premium was 6.3%. Interestingly, in his 2023 survey of 80 countries, the median market risk premium for South Africa had increased to 7.9% and the mean market risk premium had increased to 8.7%. Survey participants in South Africa appear to have become more risk averse.

If we summarise the above estimates, then the market risk premium is between 4.6% and 8.7%. We consider that the historical market risk premium of 5.1% indicated in Table 7.10, the average of 5.4% indicated by the Correia and Cramer (2008) study, as well as the range of 5.2% to 6.7% indicated by the 2022 PwC survey are closer to an appropriate market risk premium for South Africa.[7]

Warren Buffett's view

Warren Buffett does not believe in using a discount rate which includes an equity risk premium and he discounts future cash flows by using the long-term Treasury bond rate. He assumes that if he analyses a company as worthy, then no risk adjustment is required, and let's face it, with his performance record, who can blame him? However, Buffett does look for a significant "margin of safety" when he compares an investment's value to its price. He effectively makes the risk adjustment at that stage. Therefore by using a risk-adjusted rate, he would be double counting for risk. Page 7-26

Betas

The final parameter relates to the estimation of betas. We should be using forwardlooking betas, but how do we observe them? It is not possible and so we use historical market returns and prices to estimate a company's beta. How far do we go back? Do we use monthly or weekly changes in prices? We do not wish to go too far back as the resulting beta may not be relevant and may no longer reflect the company as it is today. Short intervals also mean that there is a risk of non-trading or thin trading which affects the reliability of betas. Thin trading is a factor for smaller listed companies and will result in an understatement of firms' true betas. Companies will usually obtain betas from firms such as Bloomberg or the African Institute of Financial Markets and Risk Management (AIFMRM) or other services. In Chapter 4, we have set out the AIFMRM and Bloomberg industry sector betas of the JSE. Please also refer to Appendix 4.1 and 4.2. Individual company betas may be unreliable and the sector beta may be used and comparison should be made to other firms. This may require unlevering and relevering betas for the effects of financial leverage. Studies have indicated that a firm's beta tends to migrate towards 1 over time and firms sometimes employ adjusted betas, which may be computed by making calculations such as (2/3 X firm beta + 1/3 × 1).

Adjustments to the Cost of Equity and WACC

In practice, adjustments may be made to the cost of equity for factors such as company size and a specific risk premium, as well as a liquidity or lack of marketability premium.

Small company risk premium

In practice, many companies will increase the CAPM cost of equity by a small company (stock) premium when evaluating small companies.[8] This recognises the additional risk of investing in small companies. Studies have indicated that small companies have generated higher returns than predicted by the CAPM approach. Smaller firms also experience a higher rate of corporate failure. We would expect that smaller firms should have higher betas to factor in this risk and so we need to be mindful not to double count for risk. However, the evidence indicates that the CAPM parameters tend to understate the expected return of investing in small companies. Further, small company shares tend to be illiquid resulting in lower betas due to lower trading activity.

Specific Risk Premium

Only systematic risk is rewarded in the CAPM framework. If shareholders are not fully diversified, then the CAPM estimate of the cost of equity will understate the company's true cost of equity. In the 2022 PwC survey, 66% of practitioners either frequently or always made an adjustment for specific risks. The PwC survey identifies specific risks to include the following factors:

- Dependence on key management
- One key customer or supplier
- Lack of a profitable track record
- Significant growth expectations and/or improvement in profitability
- Start-ups
- Turnaround businesses.
- Regulatory uncertainty

Page 7-27 The range of adjustments to the cost of equity based on averages made for specific risk factors was between 1.8% and 3.8%.

Lack of marketability

Investors in unlisted and private companies will require a higher required return due to the lack of marketability or liquidity. However, normally the adjustment occurs by deducting a *lack of marketability discount* after we have determined the value of an investment. We did cover this discount in Chapter 6 on Valuations.

Country risk premium

The country risk premium is captured within the yield of SA government bonds such as the R209. There is a liquid market for government bonds in South Africa and the CAPM cost of equity is based on the risk-free rate which includes a country risk premium. Therefore, for South Africa, it is less likely that we need to make further adjustments for a country risk premium.

International projects

It is important that firms estimate different required returns for different divisions and international projects, rather than use one overall WACC. For example, Sasol previously used a WACC of 13.5% for growth investments in South Africa and used a WACC of 8% to evaluate projects in the USA. The risk-free rates and betas will differ based on the sector and location of a project. In particular, differences in the risk-free rate will affect required returns. In July 2023, the risk-free rate in South Africa was 10.5% whilst it was only 3.8% in the USA.

Non-diversified shareholders

An owner of a private company will often also have a substantial amount of his/her wealth tied up in the company and therefore does not hold a diversified portfolio. This means that total risk may be more relevant than systematic risk and we may need to increase the CAPM cost of equity for this reason. Damodaran (2006)[9] indicates that a premium can be added to the cost of equity to reflect this focus on total risk rather than on market risk. He also suggests the use of a total beta, which he defines as:

$$\text{total beta} = \frac{\text{market beta}}{\sqrt{R^2}} \qquad \textit{(Formula 7.8)}$$

Effectively, we are dividing the CAPM beta by the correlation co-efficient.[10] If the CAPM beta of a company is 0.80 and the R^2 is 0.30, then the company's total beta

would be determined to be:

The resources sector and betas $$\text{total beta} = \frac{0.80}{\sqrt{0.30}} = 1.46$$

The resources sector forms over 20% of the JSE and generally companies in the resources sector will have high betas. This results in companies in other sectors having lower betas than would normally be the case if we employ the ALSI as the appropriate index. Resources should normally constitute about 10% of the market index. This means that companies in the financial and industrial sectors, including retail, will have betas that may be lower than we would expect and this will tend to understate the cost of equity if we use the ALSI. Although some analysts will use the FINDI index to determine company betas, generally companies use betas based on the ALSI and then may make adjustments either to the beta or add a premium to the cost of equity for companies in sectors outside the resources sector. For resource companies, their betas may Page 7-28 be relatively low using the ALSI and it may be argued that we should increase their reported betas. AIFMRM disclose both betas based on the ALSI and FINDI indexes.

The financial crisis, emerging markets and the cost of capital

Sasol increased its cost of capital due to the financial crisis. Other companies did the same. South African firms have on average low debt–equity ratios and this resulted in most South African firms being less affected by the financial crisis than firms in the USA and the UK. However, we need to consider the fact that the risk of investing in emerging markets has grown, and this is reflected in higher market volatility and lower sovereign risk ratings. Lower economic growth prospects, as well as budget deficits resulted in Standard & Poor downgrading South Africa's sovereign credit rating in June 2014 to BBB–, which is the lowest investment grade possible. There was a further rating downgrade to BB– for SA rand debt which will increase borrowing costs. Therefore, for the moment, we need to be cautious about the risk premium.

Taxation

The calculation of a company's WACC is after company tax but before any personal income tax. This is because investors will have varying marginal tax rates and in terms of measuring relative returns, bond returns are also subject to personal income tax. We

determine the market risk premium relative to nominal bond returns. If we used an after-tax rate, then the market premium would increase. Investors have varying personal tax rates. For example, pension funds in South Africa are taxed at a much lower rate than companies and capital gains tax rates will also vary amongst investors. In the PwC survey, respondents did not adjust the risk-free rate in the CAPM for the effect of taxation.

South Africa has a dividend withholding tax (DWT) of 20%. This is a tax on shareholders rather than a corporate tax. Tax complexities relating to the effect of dividends as well as capital gains tax add a further dimension to an already potent mix of possibilities. We have tended in this textbook to take into account only corporate tax in evaluating projects and valuing companies. This is in line with what most firms do in practice.

The role of hurdle rates

Many companies often determine the WACC, based on a CAPM approach, to work out the cost of equity but will then add a premium to WACC or multiply the WACC by a factor to establish a "hurdle" rate to discount future cash flows. Again, this implies that CAPM is not complete and companies require a higher return.

FROM THE REAL WORLD...THE COST OF CAPITAL OF SOUTH AFRICAN FIRMS

South African listed companies generally report that they will manage their capital structure in order to reduce their cost of capital. What they generally do not disclose, is what their weighted- average cost of capital (WACC) is. However, there are some companies that do disclose their WACC.

Truworths is a company that reports detailed information on the company's WACC in relation to the company's return on invested capital. This is presented in Figure 7.4.

In the period 2000 to 2008, Truworths was able to increase its ROIC at the same time that its WACC was declining due to falling interest rates. From 2008 until 2015, although ROIC started falling, it remained significantly above the group's WACC. In 2016, Truworths acquired Office, a footwear retailer in the UK, and this resulted in a large increase in its asset base and its return on invested capital (ROIC) fell from 24% to 17%. In 2018, its total assets totalled R15.5bn and operating profit was R3.9bn. Prior to the Office acquisition in 2016, its total assets were R9.1bn and operating profit was R3.4bn. The profit of Office in 2018 was R200m and total assets were over R6bn. Truworths is a superbly managed group but this again indicates the challenges of ensuring that acquisitions generate a ROIC that exceeds WACC. Truworths' ROIC in 2021 recovered and then reached a historical high in

2022. This was due to prior impairment charges on its Office UK investment thereby reducing its asset base, the closure of unprofitable Office UK stores, the renegotiation of leases, a strong post-pandemic recovery, an improvement in Office's sales and margins, movements in the Rand exchange rate, the integration of Truworths' systems and processes within the Office UK operations as well as excellent results from its other operations.

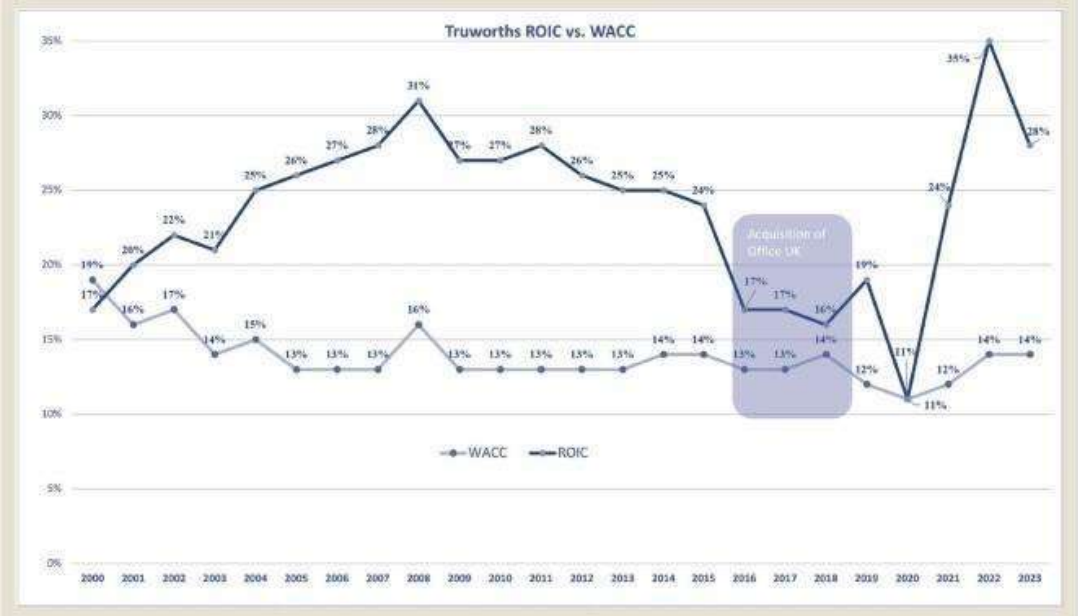


Figure 7.4: The WACC and Return on Invested Capital (ROIC) of Truworths

Woolworths states that its objective is to achieve a ROCE (Return on Capital Employed) that is never less than WACC + 5%.

Adcock Ingram reported the following in its 2022 annual financial statements:

> Discount rates reflect management's estimate of the risks. This is the benchmark used by management to assess operating performance and to evaluate future investment proposals. The discount rate is derived from the Group's weighted average cost of capital (WACC). The WACC takes into account both the cost of debt and of equity. The cost of equity is derived from the expected return on investment by the Group. The discount rate applied to cash flow projections, is 13.50% (2021: 12.60%).

Prior to its acquisition by DP World in 2022, **Imperial Logistics** reported a WACC of 11.5%. However, its hurdle rate for investments was WACC + 3%.

NASPERS reported the following in its 2023 integrated annual report:

> The discount rate used is a weighted average cost of capital. The weighted average cost of capital takes into account the cost of equity and cost of debt. The cost of equity is based on a risk-free rate adjusted for specific risks such as a country risk and equity risk premium. The cost of debt is based on the pre-tax cost of debt adjusted with a sovereign spread premium net of tax. Discount rates used ranged between 12% to 15%. Page 7-30

Discovery reports the following in its 2022 annual financial statements:

> The Discovery Limited embedded value is calculated based on a risk discount rate using the CAPM approach with specific reference to the Discovery beta coefficient. The assumed beta is fixed at 0.75. This has been set such that the risk discount rate proxies the result of a Weighted Average Cost of Capital approach with reference to the capital structure of the Group and the observed beta calculated using daily returns over a long time period. The observed beta is calculated with reference to the ALSI. The assumed beta will only change if the capital structure of the Group and/or the observed beta calculated using daily returns over a long time period suggest the beta assumption should depart significantly from the assumption at the financial year-end. As beta values reflect the historic performance of share prices relative to the market, they may not allow fully for non-market related and non-financial risk. Investors may want to form their own view on an appropriate allowance for these risks which have not been modelled explicitly.

Pick n Pay in 2022 reported its WACC to be 10.3% which compares to a WACC of 10.6% in the prior year. Pick n Pay defines WACC as follows:

> WACC is the average after tax cost of the Group's debt funding, which includes noncurrent borrowings and current liabilities, and the Group's equity funding, with each source of funding included on a proportional basis.

Interestingly, Pick n Pay discloses its WACC, excluding IFRS 16 (leases), to be 13.4% in 2023 and 14.0% in the prior year. The impact of IFRS 16 would be to increase the *reported* level of gearing of Pick n Pay and increase the proportion of reported debt in its capital structure. As the after-tax cost of debt is lower than the cost of equity, Pick n Pay may be estimating that this would reduce its WACC. Our view is that it is more likely that Pick n Pay's *actual* gearing which reflects its leases even prior to IFRS 16 and not only its *reported* gearing, (which increases when leases are reported as debt in terms of IFRS 16), is relevant to determine the company's WACC. Our view is that a firm's beta and WACC would probably already reflect the firm's actual level of financial commitments whether such commitments are recorded as debt or not. This may mean that prior determinations of the WACC may be overstated as the level of gearing was understated in a pre-IFRS 16 world. In our view, it is best to focus on the post-IFRS 16 determination of WACC. As IFRS 16 has now been in force for over 5 years, the comparability issues should be a thing of the past.

York Timbers states in its 2022 Integrated Annual Report that its WACC is 15.27% as compared to 13.61% in the prior year and states the following in relation to the determination of the firm's beta:

> In determining the weighted average cost of capital (WACC), a comparable group of forestry companies' Beta is used to determine the Beta applied in the WACC. York applied the debt/equity ratio of market participants included in its comparable company basket.

Key assumptions used by York Timbers in 2022					
Beta factor[1]	1.16	Cost of equity	18.17%	After-tax WACC	15.27%
Risk-free rate[2]	11.00%	Pre-tax cost of debt	8.25%	Debt/equity ratio[1]	24:76

[1] Relevered beta & debt/equity ratio based on a comparable company basket.[2] Based on the Government Bond 10-year yield curve

Note: Why is the cost of debt< Risk-free rate? York's long-term borrowings are based on the prime O/D rate [8.25% on 30/6/2022].

Sasol reports in its 2022 annual financial statements that its WACC was 14.41% whilst for the prior year it was 14.03%. One of the Sasol's key objectives is to deliver a ROIC for the portfolio well above WACC. Page 7-31

SUMMARY

The *cost of capital* of a company is the rate that must be earned in order to satisfy the combined required rates of return of the firm's providers of capital. Two important assumptions were made in this chapter in order to develop the procedure for establishing the weighted-average cost of capital for a company. Firstly, it was assumed that there is an *optimal capital structure* for each company and that financial managers set this finance mix as their target over the longer term. Secondly, it was assumed that *investment projects are of the same risk class as the company as a whole*, thus enabling the use of a single cost of capital for assessing all investment projects.

Three steps were identified in order to determine the WACC. They were, firstly, the *establishment of component costs*, secondly, the *weighting of each component* and, finally, the *calculation of the WACC*.

The relevant formulae for determining the cost of each component in the capital structure are based on the principles of valuation discussed in Chapter 6. The procedure for establishing weightings for each component is dependent on the target capital structure.

As a firm may have segments operating in different sectors, it may be necessary to determine divisional costs of capital which may require the unlevering and relevering of betas. It is also important to examine the practical issues involved in estimating the CAPM parameters.

This chapter has focused mainly on determining the weighted-average cost of capital of a firm. In Chapter 10, we include a section on adjusting a firm's WACC to reflect project risk.

By Johnathan Dillon M.Com CA(SA)

What does the market value of equity equate to in book value terms?

When the WACC of a company is calculated using market values then the market value of equity is calculated as the number of shares in issue (net of treasury shares) multiplied by the current share price (that is, the market capitalisation of the entity). Treasury shares are to be deducted as they respresent shares repurchased by the company and are not entitled to dividends or distributions. The company could cancel these shares if it wishes or can reissue these shares at a future date. In South Africa a company is prohibited by the Companies Act from holding its own shares and therefore any shares repurchased must be cancelled. However, in a group situation, the Companies Act allows a subsidiary to purchase up to 10% of the shares of its holding company. Again, these treasury shares (from a group perspective) are not included as they are not held "outside" of the group. (Refer to Chapter 16 for further discussion on this issue).

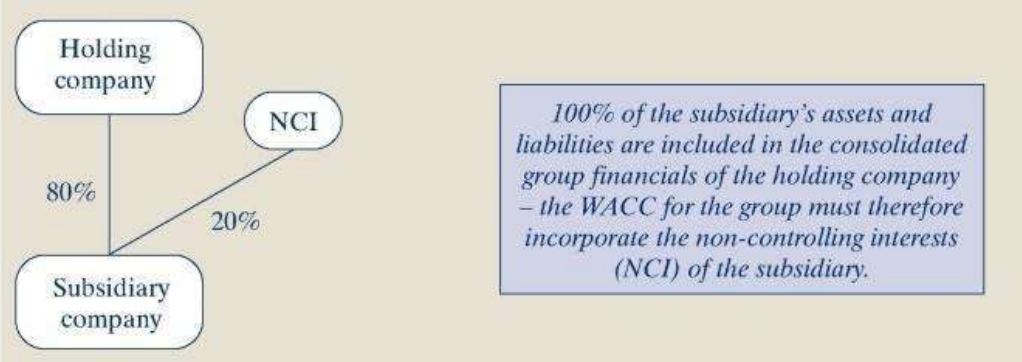
When comparing this to the book value of equity, the market value for equity is essentially the market value of the entire equity capital of the company, that is, it is the market value for share capital plus share premium, if any, plus all other reserves such as retained earnings and revaluation reserves. This is due to the fact that an investor who purchases ordinary shares in a company, at market value, is not only entitled to the original share capital but rather to his/her share of the entire equity capital, including the reserves, of the company.[11] However, as explained below, further consideration needs to be given to non-controlling interests (NCI) in a group situation. Page 7-32

How should non-controlling interests (NCI) be treated in the WACC calculation?

If the WACC for a stand-alone company is calculated using *book values* then the equity weight would be calculated by adding share capital, share premium, if any, and all other reserves as explained above. However, if the WACC is to be calculated for a consolidated group of companies then the calculation weighting for equity, based on book values, is to include noncontrolling interests (NCI). This is necessary as NCI own a portion of the subsidiary that has been consolidated into the group and the WACC calculation determines the required return on all investments (capital projects) regardless of whether the capital investment is in the holding company or the subsidiary.

If the WACC for a group of companies is to be calculated using *market values* then the equity weighting for the holding company's equity will be calculated as explained above (i.e. market capitalisation); however, NCI must still be included in the WACC calculation based on the market value of their interest in the subsidiary (that is, number of shares held in subsidiary multiplied by the subsidiary's current share price). The market capitalisation of the holding company does not include NCI as it is based on the holding company's number of

shares in issue multiplied by the market price per share. Generally the market value of NCI cannot be determined easily, in which case the book value thereof is often used as a proxy for market value. This is not an issue if NCI is immaterial but it is otherwise advisable to estimate the market value of NCI. If this is not possible based on the available information, then it may be useful to apply the holding company's price-to-book ratio to the NCI book value in order to determine its market value (unless the particular subsidiary is in a different sector). An interesting example of NCI applies to Kumba Iron Ore (see Chapter 6 for further details).



The cost of NCI equity capital included in the WACC calculation is generally assumed to equal the cost of equity of the holding company. This is a safe assumption when the subsidiary conducts a significant portion of the group's operations or the subsidiary is engaged in the same business as the remainder of the group. However, where this is not the case then the cost of NCI equity capital should be adjusted for specific differences in risk between the holding company and the subsidiary. Page 7-33

APPENDIX 7.1: CAPM AND THE FAMA-FRENCH THREE-FACTOR MODEL

Does CAPM work?

The fact that almost 100% of respondents to the PwC Valuation Methodology Survey use CAPM to determine the cost of equity seems to indicate that there is significant acceptance of the validity of CAPM to explain share returns. Is this faith misplaced? Does CAPM actually work?

If we are using CAPM to determine a company's cost of capital, then we should be confident that it reflects the real world and that share returns on average reflect CAPM. Yet there are anomalies. Firstly, CAPM indicates that a share's return depends only on its sensitivity to the market - its beta. Yet, actual returns do not reflect this and the security market line seems to be flatter than what we would predict using CAPM. This indicates that CAPM may understate the returns of low beta companies and overstate the returns of high beta companies.

This is indicated in Figure 7.5, where the risk-free rate is 8% and the average cost of capital for a company with a beta of one should be 13.5%. The empirical relationship

indicates higher returns than predicted by CAPM for low beta companies and lower returns than predicted by CAPM for high beta companies. In their study, Fama-French found that the predicted return on the lowest beta portfolio was 8.3% but the realised return was 11.1%. The predicted return on the highest beta portfolio was 16.8%, while the actual return was only 13.7%. In terms of CAPM, differences in returns should only be explained by differences in betas. In order to compensate for this and the fact that betas seem to move to one over time, companies such as Bloomberg compute what is called **fundamental betas**, which will increase low betas and reduce high betas and shift both high and low betas closer to one.

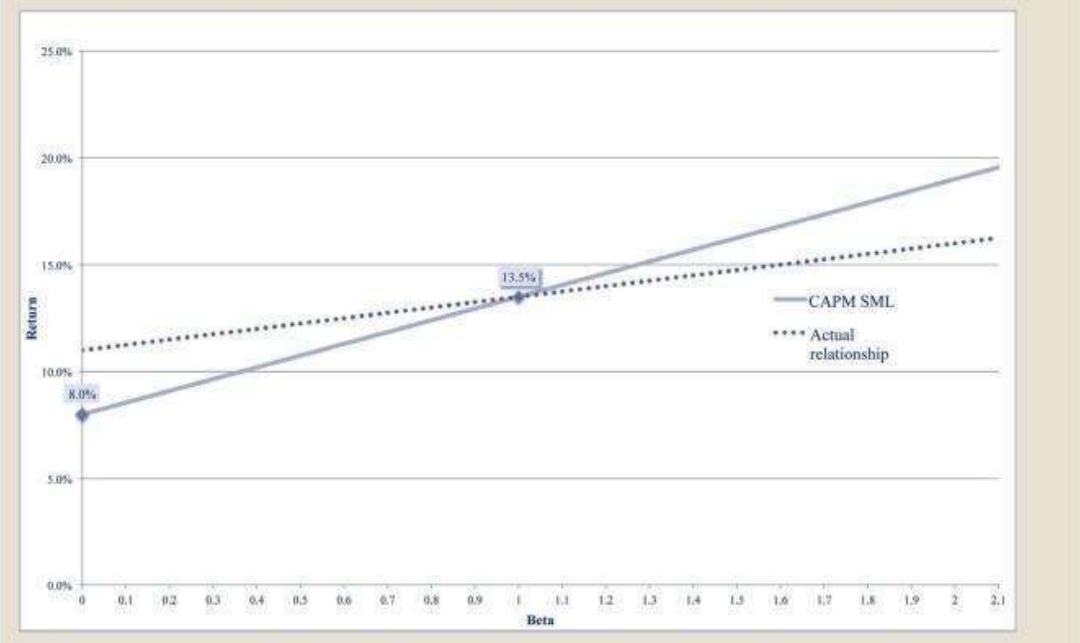


Figure 7.5: The CAPM SML and the empirical relationship of betas and returns

The following findings have arisen from studies on share returns:

- The returns of low P/E shares are higher than predicted by CAPM.
- The returns of small companies are higher than predicted by CAPM.
- High debt–equity companies generate returns higher than the returns predicted by CAPM.
- Companies with high book-to-market ratios (low price-to-book ratios) earn average returns that are higher than the returns predicted by CAPM. Page 7-34

Fama-French Three-Factor Model

Fama-French (1992) was able to show empirically that the book-to-market value per share, which reflects "value" and the market capitalisation of a company which reflects "size" are better predictors of future share returns than a company's beta. Generally, value stocks and small companies tend to produce higher returns over time. For Fama-French, these represent risk factors. Small companies are more risky and companies that have high book values per

share relative to their share price are often representative of companies that have fallen on hard times. A high book-to-market ratio may reflect financial distress. Investors may lose confidence in the company and sell shares in the market.

Fama-French produced research that indicates that a three-factor model is able to explain share returns effectively. One factor is the sensitivity to market volatility, which is CAPM, another factor relates to size (small minus big) and the final factor relates to value (high minus low). Therefore, the cost of equity is a function of the risk free rate, the market premium, the size premium and the value premium. The three-factor model was able to capture all the other effects such as the higher returns of low P/E shares.

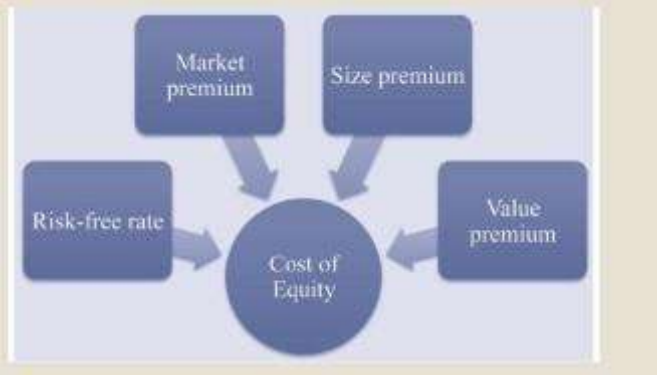


Figure 7.6: The cost of equity according to Fama-French

The **Fama-French three-factor model** is more formally described as follows:

$$r_A = r_f + \beta_A (r_m - r_f) + s_A SMB + h_A HML \qquad \textit{(Formula 7.9)}$$

Where $(r_m - r_f)$ is the market risk premium that we use in CAPM, and β_A is the beta or sensitivity of the return to movements in the market index, SMB (small minus big) represents the difference in returns between small companies and big companies and HML (high minus low) represents the difference in returns between companies with high book-to-market ratios and companies with low book-to-market ratios. The regression coefficients for these factors are indicated by s_A and h_A and we could call these factors the size beta and the value beta. Fama-French found that the three-factor model explained 90% of the variation in share returns, while CAPM often explains about 70% of the variation in share returns.

The mechanics of applying the Fama-French model is as follows: A company is placed into the small category if its market capitalisation is in the bottom half of all companies or it will be placed into the big category if its market capitalisation is in the top half of all companies. Two value-weighted portfolios are then created, small and big, and the difference in returns is then measured over time. For example, the monthly return in June for the “small” portfolio may have been 1.0% and the return for the “big” portfolio may have been 0.6%. We would then record SMB as 0.4% for the month. We would then create a series of SMB returns.

The HML returns are determined by placing a company in the "high" category if its book-to-market ratio is in the top 30% of book-to-market ratios for all companies, and the company will be placed in the "low" category if its book-to-market ratio is in the bottom 30% of book-to-market ratios for all companies. The differences in returns between the "high" and "low" categories are measured per time period. We can then regress a company's returns against these series to determine the factors s_A and h_A and we can use the series of returns to estimate the average SML and HML. Page 7-35

Fama-French reported that from 1927 to 2012, the annual size and value premiums for the USA were 3.6% and 4.8% respectively.[12]

Let's do an example. We will assume that XYZ Ltd has a beta of 1.1 and the estimated market (equity) risk premium is 5.5%. The average size premium is 3.6% and the average value premium is 4.8%. On the basis of past returns, XYZ has a size beta of 0.60 and a value beta of 0.30. This means that we can compute the cost of equity of XYZ to be 17.7% as follows:

Example 7.9: Fama-French cost of equity

		Cost of Equity
Risk-free rate		8.0%
Market (equity) risk premium	5.5%	
Beta	1.1	6.1%
Size premium	3.6%	
Size beta (s)	0.6	2.2%
Value premium	4.8%	
Value beta (h)	0.3	1.4%
Cost of Equity		17.7%

The Fama-French results have been confirmed generally in other countries, including South Africa. Of course, this could also mean that equity markets are not efficient. However, Fama-French are of the view that these reflect risk factors.

Is the Fama-French model relevant? The current practice in South Africa indicates that practitioners do not use the Fama-French model but do make adjustments at least for the size premium and also make specific risk adjustments. This indicates that the adjustments that practitioners are making will tend to increase the WACC. We may see a greater use of Fama-French in the future.[13] For example, the Chartered Financial Analysts (CFA) Institute requires that students and practitioners who aim to achieve the CFA qualification need to understand how to employ the Fama-French model to calculate a company's cost of equity.

Although CAPM is a simple and elegant way of determining a company's cost of equity, we are able to improve it by using the Fama-French three-factor model or even perhaps by making *ad-hoc* specific adjustments for the size and value factors.

Other factor models

Studies have indicated that shares that have performed strongly in the past will continue to perform strongly in the future.[14] Shares that have performed poorly in the past will continue to perform poorly in the future. Essentially, there are momentum trends in equity returns. Therefore, one is able to outperform the market by investing in shares that have performed strongly in the last 12 months and going short on shares that have performed poorly in the last 12 months. If you wish to outperform the market, buy the winners and short the losers. Momentum works not only in the USA but in markets around the world. Behavioural reasons for momentum may relate to investor herding effects and the confirmation bias. The Carhart Four-Factor Model includes a momentum factor as follows:

$$r_A = r_f + \beta_A (r_M - r_f) + s_A SMB + h_A HML + w_A WML \qquad \textit{(Formula 7.9)}$$

where w_A is the momentum beta or factor and WML is the return on a portfolio which is made of winners minus losers based on the previous 12 months of price performance. Page 7-36

Recently, Fama and French have introduced a five-factor model and the two additional factors relate to corporate profitability and investment. Companies with a high operating profitability generate higher returns and companies with a lower growth in total assets also generate higher returns. These two factors refer to the quality of a company. Fama and French have co-opted the results of other research studies to bring in these factors within an extended factor model. Investors do well by investing in highly profitable companies with low asset growth. Another factor which has been found to result in higher returns is to invest in low-volatility or low beta shares.

This has led to factor investing strategies which are based on factors which include value, small caps, momentum, low volatility, high dividend yield and quality (indicated by such measures as high profitability, low debt, accruals, cash flows, ROE and ROIC, earnings growth, dividend growth, and quality of management).

APPENDIX 7.2: THE DOWNSIDE CAPM: A ROBUST ALTERNATE TO THE CAPM?

Kwasi Okyere-Boakye BBusSci (Hons) BCom(Hons) MCom CA(SA) CFA Kwasi is an investment banking professional with a strong focus on facilitating corporate access to

diverse debt solutions across the debt spectrum, crucially supporting their balance sheet and liquidity requirements. With extensive experience in dealing with corporations both in South Africa and the broader African region, Kwasi possesses experience in the Technology, Media, Telecommunications, and Manufacturing sectors. Kwasi's versatility extends beyond traditional corporates, as he has successfully navigated complex transactions involving insurance companies, asset and investment managers, securitisation structures, and specialised finance entities.

The estimation of cost of capital (WACC) is important and has significant financial implications. Investment appraisals and valuations estimate future cash flows and discount them at a rate commensurate of the cash flow risk profile. The risks embedded in various sources of funds used by a company are matched with their relative capital structure compositions in a company to calculate a single discount or hurdle rate which is the WACC. Misestimating WACC may lead to sub-optimal investment decisions and in severe cases lead to the destruction of wealth. Equity interests, compared to other sources of funding is the riskiest; consequently, the estimation of cost of equity is probably the most elusive to measure accurately. Since its introduction to financial markets and still in modern times the capital asset pricing model (CAPM) has been widely adopted as the preferred model to capture the risk and consequently estimate the required return associated with equity ownership.

The CAPM's theoretical construct has, however, been criticised for departing from reality. The CAPM argues that the rational investor displays mean–variance behaviour (MVB) to maximise investment utility through portfolios optimisation. A well-diversified portfolio reaches an equilibrium state where all asset-specific risks have been eliminated leaving the investor exposed only to systematic risk. The systematic risk is measured by beta which is the sensitivity of a security's return against the market portfolio; beta is submitted as the best and only measure of risk and should be the only basis of rewarding investors. Across various world markets, however, the CAPM has failed empirically to explain variations in share returns and to account for market anomalies. On the JSE Securities Exchange (JSE), several authors have exposed the inadequacies of beta and thus the single factor CAPM.

In cognitive psychology, prospect theory unearths the true investor behaviour and argues that investors are loss-averse and not risk averse; that is to say that they try to minimise losses rather than minimise risks. The theory states that a given investor will display different levels of loss aversion relative to a reference point. The MVB framework values gains and losses equally; in contrast investors under prospect theory ascribe different values to gains and losses.

Furthermore, the CAPM assumes that returns in a market are normally distributed. From a market structure point of view, therefore, the CAPM is appropriate for forecasting market

 expectations in mature markets where return distributions are close to normal.

Emerging markets, on the other hand, are fragmented, they suffer from market-wide information asymmetries, their returns have been found to be non-normally distributed and are asymmetric.

The weaknesses of CAPM when investor behaviour is considered and the violations of symmetric returns distributions in certain markets have encouraged researchers to explore the mean-semivariance behaviour (MSB) framework. In the MSB framework investors pay more attention to downside risk which is the potential to sustain losses as a result of market changes. Instead of standard deviation, investors use semi-deviation as a measure of total risk. Semideviation ascribes weights only to declines in returns below a chosen benchmark and is calculated using the following formula:

$$S_i = \sqrt{(1/T)\sum_{t=1}^{T} Min[(R_t - B), 0]^2} \quad (1)$$

where R denotes returns, t indexes time, T is the number of returns in the sample. B represents the return on a chosen benchmark which may be independent of the choice of portfolio so that returns above B are not relevant, but returns below are given weight.

The equilibrium form of the semi-deviation is the downside-beta which is the alternate to the conventional beta and is determined as follows for company i:

$$\beta_i^D = \frac{Cosemivar\,(R_i, R_M)}{Semivar\,(R_M)} = \frac{\Sigma\{Min[(R_i - \mu_i), 0]Min[(R_M - \mu_M), 0]\}}{\Sigma\{Min[(R_M - \mu_M), 0]^2\}} \quad (2)$$

where R_M is the return on the market portfolio.

Under the MSB framework, the downside-beta now replaces the standard beta in the CAPM equation to derive the D-CAPM equation, which is:

$$E(R_i^D) = R_f + \beta_i^D(R_m - R_f) \quad (3)$$

To establish whether the D-CAPM outperforms the CAPM from a South African perspective, Okyere-Boakye and O'Malley (2016)[15] conducted a comparative study using 3 152 company years of 311 companies listed on the JSE between 1999 and 2014. The FTSE/JSE All Share Index was chosen as the market proxy and the benchmark return for each sample company was set at zero. Using panel data regression analysis, the study made the following findings:

1. There was statistical support for the validity of semi-variance and downside-beta, however, their explanatory powers were weaker than the low R-squares of standard deviation and beta respectively.
2. The cost of equity generated by the D-CAPM is significantly higher than CAPM-generated required returns by about 2%. This difference would significantly impact the

direction of an investment decision. This finding could be an indication that D-CAPM overestimates systematic risks whiles the CAPM suffers from underestimation.

The difference in return estimates between D-CAPM and CAPM is difficult to ignore in spite of the CAPM proving to have a slight edge over D-CAPM in the South African market. The choice between the D-CAPM and the CAPM should be one of professional judgement. They could be complimentary inputs to the investment decision. Page 7-38

SELF-STUDY PROBLEMS

S7.1

Lexton Ltd has an equity beta of 1.10. The market risk premium in South Africa is expected to be 5% and the yield on government bonds is currently 7.5%. Lexton has issued bonds and its R100 par-value bond is currently trading at R94.50. The coupon rate is 8%. The maturity date is in five years' time and the corporate tax rate is 27%. Interest is payable annually in arrears. The company has just paid the coupon interest for the current year.

Required:

(a) What is Lexton's cost of equity, based on CAPM?

(b) What is the after-tax cost of debt?

(c) Lexton paid a dividend of R0.12 per share and the dividend per share is expected to grow at 7% indefinitely. The company's share price is R2.30. What is the company's cost of equity if we use the dividend growth model?

(d) What is the weighted-average cost of capital (WACC) if the target debt–equity ratio is 50%? (Use cost of equity as per CAPM.)

S7.2

Rollon Ltd issued 0.5m corporate bonds with a par value of R100 at par and a coupon rate of 11%. The total Statement of Financial Position value of the bonds is R50m. The company's bonds are currently trading at a price of R103. Interest is payable annually in arrears. The maturity date is in four years' time. The company has a target capital structure of 25% debt, 15% redeemable preference shares, 10% non-redeemable preference shares, and 50% in ordinary equity financing. Retained earnings and contributed capital amount to R100m. This reflects the current book values of the various sources of financing.

The company has two types of preference shares in issue. There are 0.2m non-redeemable preference shares which were issued at R100 per share and there are 0.3m redeemable shares which were also issued at R100 and are redeemable at R100 per share. The maturity date of redeemable preference shares is in four years' time. Preference dividends are payable annually in arrears for both issues. Non-redeemable preference shares are currently priced at R107 and the redeemable preference shares are currently priced at R104. The coupon rates are 9% for each issue and coupon payments were recently paid.

The company's beta is 1.20 and the risk-free rate is 8%. The market premium is expected to be 5.5%. The corporate tax rate is 27%.

Required:

(a) What is Rollon's after-tax cost of debt?

(b) What is the cost of the two types of preference shares?

(c) What is the cost of equity?

(d) What is the company's weighted average cost of capital?

S7.3

Disks Ltd is a producer of music equipment. The company has identified five possible markets in respect of music preferences and has forecast their rates of return. They are as follows: Page 7-39

Market	Rate of return (%)
Popular	15%
Blues	14%
Mood music	13%
Country-Western	10%
Light classic	9%

The company has designed special sound systems to enhance the qualities of each type of music. The cost of each system is R2.0m and each is a non-divisible investment.

The current capital structure of Disks Ltd is considered to be optimal and will be maintained. At the end of the financial year the capital structure was:

Capital structure	
Debentures (10%) R1 000 par value	30%
Preference shares (8%) R100 issue price	10%
Ordinary shares (1 million issued)	30%
Retained earnings	30%

The company has a history and a stated policy of distributing 25% of its after-tax earnings as dividends. After-tax earnings were R4.00 per share, and estimated after-tax earnings for the next year are expected to reflect the historical growth of 10% p.a. The tax rate for the company is 27%.

The company is planning to finance the investment proposals by means of retained earnings and other sources of finance. In this regard, it has been ascertained that new shares could be issued at the current market price of R50 per share; flotation costs would be 10% of the current market price. Furthermore, the company can raise R3.0m from the issue of non- redeemable debentures which have a par value of R1 000 and a coupon rate of 8%, with no issue costs. The market-related interest rate for debentures in a similar risk class is 11%. The debentures would be issued at a discount to par value.

It is estimated that a total amount of R5.0m could be raised from the issue of 10% preference shares and issue costs would be negligible. Preference shares of a similar risk class are presently offering yields of 10% p.a.

Required:

(a) Calculate the amount of retained earnings available to assist in the financing of the systems.

(b) Calculate the cost of each capital component. Use the dividend yield and growth model to determine the cost of equity.

(c) Determine the breaks in the weighted-average cost of capital schedule.

(d) Calculate the weighted-average cost of capital between the break(s).

(e) Indicate which system(s) should be chosen as an investment proposal and the total capital budget.

(f) Assume the current government 10-year bond yield is 7.5% and the market risk premium is 5%. The company's beta is 1.05. What would be the cost of equity if the CAPM approach is employed? Page 7-40

Solutions to self-study problems

S7.1 Solution

1. What is Lexton's cost of equity?

The cost of equity using the Capital Asset Pricing Model (CAPM) will be as follows;

$K_e = R_f + \text{Beta (Market Risk Premium)}$

Risk-free rate			7.50%
	Beta	MRP	
Beta x Market Risk Premium	1.1	5%	5.50%
			13.00%

2. What is Lexton's after-tax cost of debt

	0	1	2	3	4	5
Cash flows	-94.5	8	8	8	8	108
Yield to Maturity (IRR)	**9.43%**					

Using financial calculator HP 10BII [Press (-94.50 PV, 5 N, 8 PMT, 100 FV). Then press I/YR for solution.]

Using the IRR function [press -94.50 CFj, 8 CFj, 8 CFj, 8 CFj, 8 Cfj, 108 CFj, then press *Shift* IRR]

	YTM	[1-tax rate]		Cost of Debt
After tax cost of debt	9.43%	73%	=	**6.88%**

3. What is Lexton's cost of equity. using the dividend growth model?

$$K_e = D_1/P_0 + g = \frac{0.1284}{2.30} + 7\% = \mathbf{12.58\%}$$

4. What is Lexton's cost of capital (WACC)?

	Cap struc.	Cost	WACC
Debt	33.33%	6.88%	2.29% [1]
Equity	66.67%	13.00%	8.67%
WACC	100.00%		**10.96%**

Debt to Equity 50.00%

S7.2 Solution

1. What is Rollon's after-tax cost of debt

	0	1	2	3	4
Cash flows	-103.00	11	11	11	111
Yield to Maturity (IRR)	**10.05%**				

Using financial calculator HP 10BII [Press (-103.00 PV, 4 N, 11 PMT, 100 FV). Then press I/YR for solution.]

Using the IRR function [press -103 CFj, 11 CFj, 11 CFj, 11 CFj, 111 CFj, then press *Shift* IRR]

	YTM	[1-tax rate]		Cost of Debt
After tax cost of debt	10.05%	73%	=	**7.34%**

2. What is Rollon's cost of preference shares?

Redeemable preference shares	0	1	2	3	4
Cash flows	-104.00	9	9	9	109
Yield to Maturity	**7.80%**				

Non-redeemable preference shares

Cost of non-redeemable preference shares = Dividend/Price

Preference dividend / Price	9.00 / 107.00	=	**8.41%**

3. What is Rollon's cost of equity?

The cost of equity using the Capital Asset Pricing Model (CAPM) will be as follows;

$K_e = R_f$ + Beta (Market Risk Premium)

Risk-free rate			8.00%
	Beta	MRP	
Beta x Market Risk Premium	1.2	5.5%	6.60%
			14.60%

4. What is Rollon's cost of capital (WACC)?

	Amount (Rm)	Proportion	Cost	WACC
Debt	50	25%	7.34%	1.83%
Prefs - redeemable	30	15%	7.80%	1.17%
Prefs - non-redeemable	20	10%	8.41%	0.84%
Equity	100	50%	14.60%	7.30%
WACC	200	100%		**11.15%**

In this case, the target capital structure reflects the book values, but this will not always be the case.

In this case, the target capital structure reflects the book values, but this will not always be the case. Page 7-42

S7.3 Solution

1. What is the retained earnings available for financing the systems?

	Current EPS	1+g	EPS	Payout	DPS	EPS-DPS
Earnings & div per share next yr	4.00	1.10	4.40	0.25	1.10	3.30
	RE per share	No. of shares	RE			
Retained earnings (RE)	3.30	1,000,000	3,300,000			

2. What is the cost of each capital component?

Cost of retained earnings						
$Ke = D_1/P_0 + g$	=	1.10 / 50.00	+	10%	=	**12.20%**
Cost of new issue of equity						
$Ke = D_1/(P_0-F) + g$	=	1.10 / 45.00	+	10%	=	**12.44%**
Cost of Debentures (after tax)		11%	x	73%	=	**8.03%**
Value per debenture	=	0.08 / 0.11	x	1,000	=	**727.27**
Cost of Preference shares	=					**10.00%**

3. What are the breaks in the weighted average cost of capital schedule?

Retained earnings	=	3,300,000 / 0.60	=	5,500,000

Co. will issue new equity if capital budget exceeds this amount.

Debentures	=	$\frac{3{,}000{,}000}{0.30}$	=	10,000,000
Preference shares	=	$\frac{5{,}000{,}000}{0.10}$	=	50,000,000

4. What is the weighted average cost of capital (WACC) between the breaks?

		Before break		After break	
Source	Weightings	Cost	WACC	Cost	WACC
Equity	60%	12.20%	7.32%	12.44%	7.47%
Debentures	30%	8.03%	2.41%	8.03%	2.41%
Preference shares	10%	10.00%	1.00%	10.00%	1.00%
			10.73%		10.88%

5. Which systems should be selected and what is the capital budget?

Select the systems that offer a return that exceeds the cost of capital	
Capital Budget for Popular, Blues & Mood music	6,000,000

6. What is the cost of equity if we use the CAPM?

The cost of equity using the Capital Asset Pricing Model (CAPM) will be as follows;

$$K_e = R_f + \text{Beta (Market Risk Premium)}$$

Risk-free rate			7.50%
	Beta	MRP	
Beta x Market Risk Premium	1.05	5%	5.25%
Cost of Equity			**12.75%**

Page 7-43 **QUESTIONS**

Question 7.1

Explain why the cost of retained earnings should be included in the cost of capital.

Question 7.2

Many business entities choose to meet financing requirements through the use of a combination of equity and debt. Provide reasons for the choice of a combination of financing instruments, rather than the use of only equity or only debt, with special reference to risk and expected return.

Question 7.3

Briefly define the term “target capital structure” and discuss the reasons why a company may set such a target.

Question 7.4

Umklomelo Ltd has a target capital structure of 60% equity and 40% debt. The after-tax cost of future debt is 5.5% and the cost of new equity is 13%. The financial

manager is currently considering a project with an expected return of 12% which will be financed from the issue of ordinary shares as all retained income is already budgeted for in more profitable projects. The company recently issued debentures and, as a result, the present capital structure is more heavily weighted towards debt.

Required:

(a) Calculate the weighted-average cost of capital.

(b) State, with reasons, whether the project under consideration should be accepted.

Question 7.5

Rex Ltd issued R30m of corporate bonds a year ago at the then ruling market yield of 11% per year. Interest rates have fallen to 9% and the company wishes to determine the cost of debt for the purposes of determining the firm's cost of capital. The corporate tax rate is 27%. Which interest rate should the firm use? Explain why. What is the market value of the bonds today if the bonds are redeemable in five years' time?

Question 7.6

It is commonly accepted that a crucial factor in the financial decisions of a company, including the evaluation of capital investment proposals, is the cost of capital. The following information is available for Umoya Ltd:

Current price per share on JSE	R2.00
Current year's dividend per share	R0.10
Expected average annual growth rate of dividends	7%
Beta co-efficient for Umoya Ltd	0.80
In addition, the following information has been established:	
Expected rate of return on risk-free securities	8%
Expected return on the market portfolio	12%

 Required:

(a) Explain what is meant by the "cost of equity capital" for a particular company.

(b) Calculate the cost of equity capital for Umoya Ltd from the data given above, using the following two alternative methods:

(i) a dividend growth model; and

(ii) the capital asset pricing model.

(c) State, for each model separately, the main simplifying assumptions made and express your opinion about whether, in view of these assumptions, the models yield results that can be used in practice.

Question 7.7

A company has a capital structure consisting of 40% debt and 60% ordinary equity. The weighted- average cost of capital is 11.5%, and equity investors require a return of 14.5% on their initial investment in 180m shares of R1. The company pays tax at a rate of 27%. If the earnings per share for the past year were 10 cents, determine the required increase in earnings before interest and tax for the current year which will increase the earnings per share to 16.5 cents.

Question 7.8

Edge Ltd is a company operating in the IT sector and the firm has a beta of 1.3. The market (equity) premium is estimated at 4% and the risk-free rate (yield on government bonds) is 7%. The company can raise debt finance at an interest rate of 9.4% per year. The corporate tax rate is 27%. The target debt–equity ratio is 33.33%. What is the firm's WACC?

Question 7.9

Sincro Ltd's capital structure is as follows:

Debt	35%
Preference shares	15%
Ordinary share equity	50%

The after-tax cost of debt is 6.5%; the cost of preference shares is 10%; and the cost of ordinary share equity is 13.5%.

As an alternative to the existing capital structure for Sincro, an outside consultant has suggested the following modifications:

Debt	60%
Preference shares	5%
Ordinary share equity	35%

Under this new and more debt-oriented arrangement, the after-tax cost of debt is 8.8%; the cost of preference shares is 11%; and the cost of equity is 15.6%.

Required:

(a) Calculate Sincro's existing weighted-average cost of capital.

(b) Recalculate Sincro's weighted-average cost of capital, using the capital structure suggested by the consultant.

(c) Discuss the issues which are pertinent to the choice between the two alternative capital structures. Page 7-45

Question 7.10

Waka Ltd has the following capital structure (R millions):

	Proportion	Rate at date of issue	Current
Long-term debt	250.0	7% average rate	11% yield
Preference shares (non-redeemable)	50.0	6% stated rate	9% yield
Ordinary shares (10 million shares)	400.0	–	Price range R45 – R60
Total capitalisation	700.0	Recent price R50	

The company's beta is currently estimated at 1.2, and the risk-free return is considered to be 7.5%. The most recent estimate of the return from the JSE overall index is 13.5%. Assume a tax rate of 27%. The long-term debt of R250m is repayable in 5 years' time.

Required:

(a) Calculate the various weighted-average costs of capital for Waka Ltd, based on original costs and marginal costs, using both a book value and a market value weighting for each case.

(b) Discuss the range of costs calculated and suggest an appropriate cost of capital, with justification.

Question 7.11

The market risk premium in South Africa is expected to be 5% and the yield on government bonds is currently 7.8%. Lappelle Ltd has an equity beta of 0.90. Lappelle has issued bonds with a par value of R100, which are currently priced at R96.00. The annual coupon rate is 9%. The maturity date is in four years' time and the corporate tax rate is 27%. Interest is payable semi-annually in arrears. The company has just paid the coupon interest for the current six month period.

Required:

(a) What is Lappelle's cost of equity, based on CAPM?

(b) What is the after-tax cost of debt?

(c) What is the weighted-average cost of capital (WACC) if the target debt–equity ratio is 25%?

Question 7.12

Asanda Ltd, a company with R100m in assets and 50% debt in its capital structure, is considering a series of projects in the same risk category as the overall business, based on a capital budget of R25m. The company's after-tax weighted-average cost of capital is 10.4%, the marginal cost of debt is 8% before-tax and the marginal tax rate is 27%.

Required:

If the project is financed exclusively with new equity, calculate the rate of return it must earn to be acceptable. Ignore flotation costs.

Question 7.13

A company with R100m in assets and 40% debt in its capital structure is considering projects totalling R20m. Its shares are currently selling for R20, with a price–earnings ratio of 10, and a dividend payout ratio of 40%. The company can raise up to R30m in long-term debt at 11%. New ordinary shares can be issued at the current price, less flotation costs of 5%, and the company intends retaining its existing capital structure. The expected growth rate in dividends is 10% per year.

Required:

Calculate the cost of capital to be compared against the return on the project if the marginal tax rate is 27% and the book value of net assets equals their market value.

Question 7.14

Smith & Nguni Ltd operate in the IT sector. The company writes and provides accounting software for about 10% of the South African market. The company expects to maintain a steady increase in earnings and dividends of 8% each year for the foreseeable future. The company's current dividend is R1.40 per share and the company's share price is currently at R42.

The company's equity beta is 1.55 and the risk-free rate is 8%. The market premium is 5.5%. The interest rate on long-term debt is currently 10%. The corporate tax rate is 27%.

Required:

(a) What is the cost of equity if the company uses the dividend growth model?

(b) What is the weighted-average cost of capital if the company has a target debt–equity ratio of 100% and the company uses the CAPM method to determine the cost of equity?

Question 7.15

Coldmax Ltd provides refrigeration services to South African food companies. The company's capital structure is presented below:

Capital Employed	R000
Long-term debt - debentures	13,244
Preference shares	1,892
Ordinary share capital	22,704
	37,840

The debentures which have a par value of R100 are currently priced at R112 and are redeemable in five years' time. The coupon rate is 10% per year. The preference shares, which had an issue price of R100, are currently priced at R118 and the preference shares offer a fixed dividend rate of 12% per year. The preference shares are non-redeemable. The company's equity beta is 1.3 and 10-year Treasury bonds are currently yielding 8%. The market premium is estimated to be 5%. The company's book value capital structure is expected to reflect the target capital structure in the future. The corporate tax rate is 27%.

Required:

What is the firm's weighted-average cost of capital (WACC)?

Question 7.16

Cax Ltd issued corporate bonds with a coupon rate of 8.4%. The face value value is R100 and the bonds are stated on the Statement of Financial Position at its total par value of R80m. The bonds are currently trading at a price of R94. Interest is payable annually in arrears. The maturity date is in four years' time. The company has also issued variable loan finance of R10m at a current interest rate of 9.2% per year.

The company has 200 000 non-redeemable preference shares which were issued at a price of R100 each. Preference dividends are payable annually in arrears. The non-redeemable preference shares are currently priced at R108. The coupon rate is 9% and the company has recently paid the preference dividend for the current year. The company's equity beta is 1.15 and the risk-free rate is 7%. The company uses a market

premium of 6.5% as this is the average between 5–8% which is the range of market premiums recommended by some analysts. The current share price is R3.20 and the company has 30 million ordinary shares in issue. Page 7-47

Required:

(a) What is Cax's after-tax cost of debt?

(b) What is the cost of the preference shares?

(c) What is the cost of equity?

(d) What is the company's cost of capital? Employ market value weights for the firm's capital structure.

(e) What is the company's WACC if the company has a target capital structure of 40% corporate bonds, 20% preference shares and 40% in ordinary equity financing?

Question 7.17

Leighton Sizwe Ltd has three divisions which operate in different industry sectors. The company is currently evaluating one project from each division. The equity betas reflect industry unlevered betas that have been relevered to reflect the company's capital structure. The company has determined the internal rates of return (IRR) of each project, which indicates the implicit return of each project. The company's debt–equity ratio is 50%. The corporate tax rate is 27%.

Division	Beta	IRR
A	1.30	14%
B	1.10	12%
C	0.80	10%

The company's overall beta is 1.10 and the current yield on government bonds is 7%. The equity or market premium is expected to be 5%. The current cost of long-term debt is 9% although the company did issue debentures three years ago at a yield of 12%.

Required:

Indicate the required return of each division and whether the company should select all the projects. How would the firm's investment decisions change, if it used a single company-wide WACC to evaluate all projects?

Question 7.18

Skosana Ltd manufactures cables to be used in the construction sector. The company uses both debt and equity to finance operations. Share capital amounts to R50m and there are 100m ordinary shares in issue. The current share price on the JSE is R1.30. The company's long-term debentures are currently yielding 9% and the book value of the debentures is R150m. The coupon rate is 11%. The company's beta is estimated to be 1.20 and the market premium is 6%. Government bonds are currently yielding 7.5% per year. The company tax rate is 27%. The debentures are redeemable in four years' time. Retained earnings are R40m.

Required:

What is the company's weighted-average cost of capital (WACC) if the company uses market value weights for its capital structure? What is the WACC if the company uses book values?

Question 7.19

The long-term capital employed and other information relevant to three companies engaged in the same type of business is as follows: Page 7-48

	R Ltd	S Ltd	T Ltd
Number of ordinary shares	3,000,000	16,000,000	110,000,000
Market price per ordinary share	0.70	2.40	0.30
EPS	0.09	0.36	0.025
Dividend per share	0.018	0.120	0.020
Growth rate in dividends	12%	10%	9%
EPS - forecast	0.100	0.400	0.025
Long term loans	-	48,250,000	26,350,000
Market price of loans (per RI 00)	-	80	60
Nominal interest rate		7.5%	5.0%

It may be assumed that the redemption dates of the long-term loans are not relevant to this question, and that the standard rate of company tax is 27%.

Required:

(a) Calculate each company's current P/E ratio, earnings yield, dividend yield, and dividend cover, and comment in detail on possible reasons for differences between the P/E ratios of the three companies.

(b) Calculate each company's cost of equity capital and comment on the results obtained and also on other methods of calculation that might have been used if additional information had been available.

(c) Calculate the weighted-average cost of capital for each company, and comment briefly on the results obtained.

(d) List three purposes for which cost of capital calculations might be needed, and state how you would define cost of capital for each purpose.

Question 7.20

At the end of 20.6, Standard Steel Products had R800m total net assets. Management desires to increase its production machinery during 20.7 by R200m in order to provide for new product lines. The increase can be effected on a partial basis, depending on funds available and the cost of those funds.

Debt financing will be raised at an 11% rate and will sell at par. Preference shares will be issued at R100 per share, less share issue expenses of R2.00 per share. The preference dividend will be 12% (based on R100). Ordinary shares, which are currently selling for R31 a share, can be sold to net R30 after flotation costs. Internal funding available from retained earnings is estimated to be R50m.

The firm has had a 5% dividend yield and a growth rate of 6% over the past few years. Its tax rate is 27%.

Standard's present capital structure listed below is considered optimal.

	Rm	Rm
Share Capital (20m shares in issue)	200	
Retained earnings	300	500
Preference shares (Im shares in issue)		100
Debentures - 8% coupon rate	100	
Debentures - 10% coupon rate	100	200
		800

Required:

(a) Determine the amount of equity capital which must be issued in order to maintain the present capital structure if the full amount of the capital budget is expended. Page 7-49

(b) Determine the amount of equity funding which must come from the sale of new ordinary shares if the full amount of the capital budget is expended.

(c) Calculate the weighted-average cost of capital that should be used in the evaluation of the expansion opportunity if it must be accepted in its entirety.

(d) Draw a graph illustrating the break which will occur in the marginal cost of capital schedule.

Question 7.21

Packline (Pty) Ltd wants to expand production capacity by 40% which will involve a R120m investment in plant and machinery. Management wants to maintain a 100% debt-to-equity ratio. The dividend policy has been to distribute 30% of their after-tax earnings which this year was R40m.

Debt financing will be at an 8% coupon rate and will sell at a discount which will result in a yield to maturity of 11%. Ordinary shares, which are currently selling for R30 a share, can be sold to net R27 after flotation costs.

The firm has a 5% dividend yield and a growth rate of 6% over the past few years. The tax rate is 27%.

The yield on the 10-year government bond is 8% whilst the yield on short-term Treasury Bills is 7%. The market risk premium is expected to be 5%. Due to liquidity issues, the company will use the sector beta adjusted for differences in financial leverage. The sector beta is 0.85 and the sector's debt–equity ratio is 25%.

Required:

(a) Calculate the appropriate cost of capital that should be used in the evaluation of the expansion opportunity if it is an indivisible project. Employ CAPM to determine the cost of equity.

(b) Assuming the expansion could take place on a proportional basis, identify the issues which the financial manager would have to address in arriving at a decision.

Question 7.22

Kei Print Ltd is expanding into the production of paper products. Point Paper Ltd is a listed company that is currently producing paper products. The equity beta of Point Paper is 1.15 and the company's debt–equity ratio is 50%. Kei Print has a debt–equity ratio of 100%. The current yield on long-term government bonds is 8% and the expected market return is 14%. Kei Print is currently able to borrow at an interest rate of 10% and the bank will finance any loan in the new sector also at 10%. The tax rate is 27%. What is the cost of capital that Kei should use to discount the cash flows from the new project?

Question 7.23

Rupab Co is a manufacturing company that wishes to evaluate an investment in new production machinery. The machinery would enable the company to satisfy increasing demand for existing products and the investment is not expected to lead to any change in the existing level of business risk of Rupab Co. The machinery will cost R25 million, payable at the start of the first year of operation, and is not expected to have any scrap value. Annual before-tax net cash flows of R6.8 million per year would be generated by the investment in each of the five years of its expected operating life. These net cash inflows are before taking account of expected inflation of 3% per year. Initial investment of R2.4 million in working capital would also be required, followed by incremental annual investment to maintain the purchasing power of working capital.

Rupab Co has in issue five million shares with a market value of R38.10 per share. The equity beta of the company is 1.2. The yield on short-term government debt is 4.5% per year and the equity risk premium is approximately 5% per year. The debt finance of Rupab Co consists of bonds with a total book value of R20 million. These bonds pay annual interest before tax of 7%. Page 7-50

The par value and market value of each bond is R100. Rupab Co pays taxation once a year in arrears at an annual rate of 25%. Capital allowances (tax-allowable depreciation) on machinery are on a straight-line basis over the life of the asset.

Required:

(a) Calculate the after-tax weighted-average cost of capital of Rupab Co.

(b) Prepare a forecast of the annual after-tax cash flows of the investment in nominal terms, and calculate and comment on its net present value.

(c) Explain how the capital asset pricing model can be used to calculate a project-specific discount rate and discuss the limitations of using the capital asset pricing model in investment appraisal.

(*ACCA Financial Management – adapted*)

Question 7.24

The financial director of Megabond Ltd is considering next year's investment projects and the financing thereof. The company is faced with the following projects, some of which are divisible, while others are not.

Projects	Amount to be invested	Nature of project	Expected rate of return
K	R75m	Indivisible	15.5%

Projects	Amount to be invested	Nature of project	Expected rate of return
L	R15m	Indivisible	10.9%
M	R20m	Divisible	12.2%

Historically, the company has had a dividend payout ratio of one third of net income available for dividends, and this trend is likely to continue for the foreseeable future, as is the growth rate in dividends. The last dividend paid was 150 cents per share. The current share price is R33.00 and, based on this price, it is expected that the dividend yield for the coming year will be 5%. Expected net earnings attributable to ordinary shareholders for the coming financial year are expected to be R45m. The company is financed solely with shareholders' equity and longterm debt, and the debt–equity ratio is 33.3%. This ratio is assumed to be optimal and will be retained. The tax rate for the company is 27%.

Should the company need to raise funds in the capital market, it will need to pay flotation costs of R1.00 per share for an issue of new equity. It can also raise R25m debentures at 10%, while any amount between R25m and R250m will attract interest of 11%.

The current yield on government bonds is 8% and the market risk premium is expected to be 5%. The company's equity beta is 1.1.

Required:

(a) Calculate the weighted-average costs of capital to be used to determine which, if any, of the three projects would be acceptable. Employ both the CAPM and the dividend growth model to determine the firm's cost of equity. Which method to determine the cost of equity is most often used by South African companies?

(b) Calculate the capital budget for the forthcoming year.

Question 7.25

Brundel Ltd is evaluating expansion opportunities with similar expected returns, which will require a total of R30m to finance. In 20.6 a dividend of R5.50 per share was paid. At present the shares are quoted at R50 per share. A dividend growth of 8% per annum is expected for the foreseeable future. The company would realise R50 per share on a new issue but would have to pay flotation costs of R4 per share. In addition, 12% preference shares or 10% debentures could be issued. Any debentures issued would be offered at face value. Retained earnings are expected to amount to R12m in 20.7.

The following summarised extract is from the 20.6 financial statements. This capital structure (based on book values) is considered optimal by the board of directors.

	(R000)
Ordinary share capital	22 000
Retained earnings	6 000
Ordinary shareholders equity	28 000
Preference share capital (10% p.a.)	8 000
Debentures (8% p.a.)	20 000
Total capital employed	56 000

You may assume a tax rate of 27%.

Required:

(a) Determine the amount which must be raised from the issue of new equity in order to finance the expansion opportunities while maintaining the current capital structure.

(b) Find the after-tax cost of 20.7 retained earnings and of new ordinary share capital if raised. Use the dividend growth model to determine the cost of equity.

(c) Assume that the current yield on long-term government bonds is 8% and the market risk premium is 6%. The firm's equity beta is 0.90. The current yield on 12-month Treasury bills is 7%. Use the CAPM to determine the firm's cost of equity.

(d) Calculate the appropriate weighted-average cost of capital that should be used in the evaluation of the expansion opportunities. Use the CAPM to determine the cost of equity.

(e) Explain whether your answer to (d) would change if a total of R60m was needed to finance the total expansion, assuming that the funds could be raised at the same rates.

Question 7.26

The directors of Hacklett Ltd are to meet to consider the company's expansion programme. They have requested you, the financial manager, to provide an estimate of the company's cost of capital as a guide in making the relevant decisions. The following data have been extracted from the latest financial statements:

Capital Employed	R000
Ordinary share capital (5m shares in issue)	10,000
Non-distributable reserves	10,000
Retained income	40,000
Ordinary shareholders funds	60,000
Preference shares (10m shares of R1 each)	10,000
Debentures	20,000

Capital Employed	R000
	90,000
Current assets	70,000
Current liabilities	40,000
Net current assets	30,000

Included in current liabilities are short-term borrowings (including overdraft) of R2.5m. The earnings per share was R3 and the dividend per share was R1.80. You have obtained the following further data from company records, consultation with the directors and other sources.

- The preference shares were issued several years ago at a rate of 7%. Investors would require a return of 9% on their investment if new shares were to be issued today; the investment bankers recommend an issue price of R5 and flotation costs would amount to R0.25 per share. The preference shares are non-redeemable. Page 7-52
- The debentures were also issued several years ago, at a rate of 8.5%. New 5-year debentures can be issued at 11% and issue costs would be negligible. The current debentures are redeemable in 5 years' time.
- While the Statement of Financial Position reflects short-term borrowings of R2.5m, you are aware that the level of usage fluctuates and can be estimated to average R10m. The interest rate on short-term borrowings is 11%.
- The compound growth rate of earnings per share over the last ten years has been 12%. From your discussions with analysts it appears that they expect this rate to be maintained. Although the company has no clearly stated dividend policy, dividend cover has remained fairly constant in recent years.
- Over the last year the share price has fluctuated between R15 and R22 but has tended to stabilise at R20. The investment bankers feel that an issue at R18 would be successful and they would be happy to attend to all matters relating to the issue at a cost of R0.50 per share.
- The current tax may be taken as 27%.
- The target capital structure is represented by the values on the Statement of Financial Position.

Required:

(a) Assuming that the directors wish to maintain the current capital structure and that they do not wish to issue further ordinary shares, calculate the weighted-average cost of capital.

(b) Calculate the cost of capital assuming that the directors:

  (i) want to buy back the preference shares on the open market and have no intention of issuing further preference shares;

  (ii) want to use short-term borrowings only as bridging finance for short periods.

(c) Using the assumptions in (b), establish the maximum capital budget before the company would be required to issue further ordinary shares.

(d) It is possible that the company's future plans could include investments in assets of a markedly different nature from those which are currently in use. It is also possible that a reduction in the dividend payout ratio of up to 40% may be required to facilitate expansion. Explain the effect of each of these possibilities on the cost of capital of Hacklett Ltd.

(e) Explain briefly how the capital asset pricing model could be used to arrive at a cost of equity for Hacklett Ltd. Assume that the yield on government bonds is 8% and the company's beta is 0.85. The market risk premium is 6%. Determine a revised WACC using CAPM and market values to reflect the target capital structure.

Question 7.27

Burse Co wishes to calculate its weighted-average cost of capital and the following information relates to the company at the current time:

Number of ordinary shares	20 million
Book value of 7% convertible debt	R290 million
Book value of 8% bank loan	R20 million
Market price of ordinary shares	R55.00 per share
Market value of convertible debt	R107.11 per R100 bond
Equity beta of Burse Co	1.2
Risk-free rate of return	4.7%
Equity risk premium	6.5%
Rate of taxation	30%

Burse Co expects share prices to rise in the future at an average rate of 6% per year. The convertible debt can be redeemed at par in eight years' time, or converted in six years' time into 1.5 shares of Burse Co per R100 bond.

 Required:

(a) Calculate the market value weighted-average cost of capital of Burse Co. State clearly any assumptions that you make.

(b) Discuss the circumstances under which the weighted-average cost of capital can be used in investment appraisal.

(c) Discuss whether the dividend growth model or the capital asset pricing model offers the better estimate of the cost of equity of a company.

(*ACCA Financial Management – adapted*)

Question 7.28

Groot Marico Breweries Ltd ("GMB") brews and sells beer in Africa, South America, Europe and China. GMB was founded over one hundred years ago in Johannesburg and has grown into the second largest brewing business globally. GMB is listed on the Johannesburg Stock Exchange (JSE) and has a secondary listing on the New York Stock Exchange (NYSE). GMB has grown mainly through acquisitions over the decades and has funded these acquisitions mostly by means of debt.

Takeover bid by Charlton Beverages Inc.

Charlton Beverages Inc. ("CBev") is the world's largest beer brewing business with operations in North and South America, Europe and Asia Pacific. CBev, like GMB, has grown mainly through acquisitions. CBev is listed on the NYSE and has recently obtained a secondary listing on the JSE. CBev's earnings before interest and taxation (EBIT) is approximately double that of GMB.

CBev made an offer on 1 November 20x6 to acquire 100% of the shares in issue of GMB at a cash price of ZAR850 per share. CBev has also extended a partial share alternative to GMB shareholders on the following terms and conditions:

- GMB shareholders electing the partial share alternative will be entitled to receive ZAR75 in cash and 0.60 CBev restricted shares for each GMB share held;
- The CBev restricted shares will have the same voting and dividend rights as CBev ordinary shares except that the restricted shares may not be sold on any recognised stock exchange for a period of five years following issue;
- Each CBev restricted share will automatically convert into one CBev ordinary share in five years' time; and

- GMB shareholders must elect to receive either cash or the partial share alternative in respect of all their GMB shares held.

CBev has obtained irrevocable undertakings from GMB shareholders holding a total of 50% of the total GMB shares in issue to vote in favour of the takeover offer. They will elect to receive cash instead of the partial share alternative. The takeover of GMB by CBev is subject to numerous suspensive conditions, including obtaining regulatory approvals from the competition authorities in every major territory in which CBev and GMB operate.

Proposed funding of the GMB takeover by CBev

CBev intends funding the largest part of the cash consideration due to GMB shareholders by raising debt. CBev's transaction advisor, ASA Investment Bank, a leading global investment banking group, has proposed that CBev issue debentures on the following key terms and conditions:

Aggregate principal amount of the debentures:	USD60 000 million
Issue price of the debentures to the public:	98% of the principal amount
Fixed annual interest rate of the debentures:	4.8%
Payment of interest on debentures:	Annually in arrears on 30 June
Redemption of the debentures:	100% of the principal amount 10 years after the issue date
Underwriting and issue costs:	0.5% of the principal amount

Page 7-54 The Board of Directors of CBev supports the use of debt to fund the GMB acquisition but would like the group to have sustainable and manageable debt levels post acquisition. CBev currently pays an average interest rate of 5.5% on short-term borrowings and 4.5% on long-term debt.

Selected historical financial information

Both GMB and CBev have June financial year ends and have the US dollar as their functional and reporting currency. Selected information from the most recently audited annual financial statements of GMB and CBev for the year ended 30 June 20x6 is summarised in the table below:

	CBev USD million	GMB USD million
Revenue	36,600	20,100
EBIT	12,800	6,400
Net finance costs	600	100
Total equity	43,700	26,200

	CBev USD million	GMB USD million
Aggregate net interest-bearing debt	12,500	2,000
Effective tax rate	20%	30%
Number of shares in issue	2,000,000,000	1,500,000,000

Selected historical market information

The share prices of GMB and CBev at selected dates together with the USD:ZAR exchange rates on those dates are summarised in the table below:

	30/9/20x6	**1/11/20x6**	**31/12/20x6**	**20/1/20x7**
GMB share price in SA Rand	675.00	800.00	830.00	845.00
CBev share price in USD	96.40	98.40	105.00	102.00
ZAR:USD exchange rate	13.50	13.00	13.80	14.20

ASA Investment Bank performs its own research on beta coefficients and market risk premiums (higher return expected from equities versus risk-free instruments). Some of its research results are summarised in the table below:

	30 September 20x6
Market risk premium	
South Africa	6.0%
USA	4.5%
Risk-free rates	
South Africa	8.5%
USA	3.5%
Unlevered beta co-efficient	0.80
Levered beta co-efficient	0.84

ASA Investment Bank uses the Hamada formula to lever and unlever beta coefficients.

Hamada formula

$$\text{Beta (levered)} = \text{Beta (unlevered)} \times [1 + [(1 - T) \times (\text{Debt/Equity})]]$$

Required:

a) Calculate the effective pre-tax interest rate of the debentures assuming that CBev issues debentures of USD60 000 million on 1 July 20x6 on the proposed terms and conditions.

b) Calculate the ZAR equivalent per share that each GMB shareholder electing the partial share alternative would have received if the purchase price had been settled on either 1 November 20x6 or on 20 January 20x7 and compare both to the cash offer.

c) Discuss the factors that South African based GMB shareholders needed to consider when they decided whether to elect the cash offer or the partial share alternative.

d) Page 7-55 Calculate CBev's weighted average cost of capital (WACC) on 30 September 20x6.

Assume the following:

- CBev had acquired 100% of GMB at that date;
- The total purchase consideration was USD94 500 million;
- USD60 000 million of the total purchase price was settled in cash;
- USD34 500 million of the total purchase price was settled by the issue of CBev restricted shares;
- CBev issued USD60 000 million of debentures to fund the cash portion of the purchase price; and
- The CBev group's effective tax rate is 20% post acquisition.

e) Analyse and discuss the CBev group's gearing levels post the acquisition of GMB. (*SAICA ITC 2017*)

Question 7.29

Plastix (Pty) Ltd ('Plastix') manufactures plastic water bottles, and the firm supplies a wide range of customers country-wide, including well-known water companies. Plastix has a 31 December financial (FY) yearend. Given the pressure to be more ESG-conscious and reduce plastic waste, Plastix has decided to start a new initiative by investing in bioplastic manufacturing. Bioplastic manufacturing uses renewable plant-based sources and demand for bioplastic products is expected to increase over time. Traditionally, petroleum has been the main ingredient in plastic bottles, making the input costs sensitive to oil prices. Plastix is financed with a mix of debt and equity and aims to achieve a debt ratio of 40% in the long term. The following information pertains to the current debt financing of Plastix:

- 12 000 non-redeemable cumulative preference shares were issued five years ago at R100 each. Dividends are paid at 10% annually in arrears. Investors would require a return of 8% on their investment if new shares were issued today. Due to supply chain shocks and related issues in financial performance, the preference dividends had not been paid for the past two financial years (FY20x1 and FY20x2). The company's performance is recovering, and it is expected that preference dividend payments will be resumed at the end of FY20x4, by which time dividends in arrears will have been fully paid.
- A loan of R5 million was obtained five years ago from the company's bank. The loan bears interest at a fixed rate of 8.5% per annum and is repayable in ten equal annual instalments in arrears, starting one year post issue. If Plastix were to apply for a similar loan today, the interest rate would be 8%.

Plastix sales revenue in 20x2 was R36.179m resulting in net profit after tax of R2.951m. The operating profit was R2.869m and depreciation was R424000. The corporate tax rate in South Africa is 27%. The following market data was available on 31 December 20x2:

- R20x3 government bond (redemption date 20x3) 5.82%
- R186 government bond (redemption date 20x6) 8.69%
- Equity market risk premium 7.00%

Information for listed entities in South Africa with similar operations to Plastix	**Average for the packaging industry**	**Average for plastic bottle manufacturers**
Earnings before interest, tax, depreciation and amortisation (EBITDA) multiple	5.8	5.3
Unlevered beta	0.85	0.91
Debt-equity ratio	50%	45%

Hamada formula: $\beta_L = \beta_U [1 + (1 - t)(D/E)]$

Required:

Determine Plastix's weighted average cost of capital as at 31 December 20x2. [*SAICA ITC 2023 (adapted – includes only the WACC section. The NPV section is in Ch8, Q8.31.*] Page 7-56

Question 7.30

OPM is a private equity company that invests in established small to medium unlisted companies that have growth potential within the agricultural and industrial sectors. OPM was required to undertake valuations of its investments as of 31 May 20x4. OPM owns a controlling interest in Crescent Chickens (Pty) Ltd, which supplies poultry products to its customers throughout the country. Its customers are major food retailers and fast food chains and the company has a 2% market share. As part of the valuation process, OPM is required to determine the WACC of Crescent Chickens (Pty) Ltd and the following information is relevant:

- The current yield on the long-term government bond is 7.8% per annum, while short-term government treasury bills are yielding 4.9% per annum.
- The prime overdraft rate is presently 8.5% per annum.
- The average historic market risk premium for the JSE equity market is 5.5% per annum.
- Current information applicable to listed poultry producers and the wider agricultural sector is presented below:

Variable	Average value of all listed poultry producers	Average agricultural sector value
Levered beta	1.58	1.31
Unlevered beta	1.26	1.01
Debt-equity ratio	35%	40%
Earnings before interest, taxation, depreciation and amortisation (EBITDA) multiple	5.5	6.7

Mr. Henry Koopman (a manager at OPM) believes the most appropriate way to lever and unlever betas is to use the following Hamada formula:

$$B_U = B_L / [1 + (1 - T)(D/E)]$$

Where: B_U = unlevered beta

B_L = levered beta

T = taxation rate

D = market value of debt

E = market value of equity

The corporate tax rate is 27%. Interest on long-term borrowings is payable at a variable rate linked to the prime overdraft rate. The interest rate payable is set at 250

basis points (2.5%) above the prime overdraft rate. OPM's target long-term debt–equity ratio for Cresent Chickens is 25% and is based on market values.

Required:

Determine the weighted average cost of capital (WACC) that should be employed to discount the future operating cash flows of Crescent Chickens (Pty) Ltd. [*SAICA ITC (adapted – this is limited to the section which requires the calculation of WACC)*]

Question 7.31

DentHealth Products (Pty) Ltd is a medium-sized family-owned company that mainly supplies dental and healthcare products to dentists and major retailers such as Clicks and Diskem. The company's founder, Gary Moult has one daughter, Janet, who has just qualified as a Chartered Accountant (SA) and has become the chief financial officer of the company. The major Page 7-57 shareholders are Gary, who has a 70% shareholding, HPE, a private equity company which owns 20% and two other individual shareholders with 5% each. The company is required to evaluate an investment project in the retail sector. Gary has always used a discount rate of 16% to determine the NPV of any project that the company has invested in over the past few years and has proposed that the company uses the same rate to evaluate this investment project. Janet is of the view that the company should use a WACC as the discount rate and should use the CAPM to determine the cost of equity. CAPM only considers the systematic risk of projects. Gary is sceptical but has asked Janet to make a presentation at the board of directors' meeting next week on the use of WACC and to estimate the WACC for the project.

Janet has undertaken the following research in preparation for the meeting and in order to determine the WACC of DentHealth, as well as the WACC that should be used to evaluate the new project which will cost R20 million. The following information is relevant: ■ The current yield on the 10-year government bond is 8%. The current one-year government

Treasury Bill rate is 6.5%.

- The market risk premium is estimated to be 5.5%.
- TotalChem is significantly larger but similar listed company to DentHealth and has an equity beta of 0.95 and a debt–equity ratio of 100%.

- A similar retail company operating in the same sector as the proposed investment project has an equity beta of 1.4 and a debt–equity ratio of 150%.
- The corporate tax rate is 27%.

DentHealth has a fixed rate 10-year loan on which it is paying an interest rate of 11%. The company can borrow on a long-term basis on a fixed rate at the 10-year government bond yield plus a spread of 400 basis points (4%). The current prime overdraft rate is 9% and the company can borrow on a short-term basis at prime plus 1%. The company can also enter into a 10-year variable rate loan at prime plus 2%.

Gary currently has a capital structure of 80% equity financing and 20% debt financing. Janet has indicated that the target capital structure should reflect a debt–equity ratio of 40%, which in her view, is reasonably conservative and believes the company should move progressively towards this target capital structure. DentHealth is highly profitable. Gary is not keen to increase the gearing of the company but Janet has received support from HPE, which had been requesting Gary to increase the financial leverage of the company for some time now. Janet has also been informed that HPE require its private equity investments to generate an internal rate of return (IRR) of 20% and HPE is looking to exit its investment in DentHealth within 10 years.

Janet needs to explain why the WACC and CAPM method is an improvement on the current practice of using a single discount rate to evaluate projects. Further, she will need to explain CAPM and the concept of systematic risk.

Required:

1. Set out some of the disadvantages of using a single discount rate to evaluate projects.
2. Determine the WACC of DentHealth, as well as the WACC that should be used to evaluate the new retail project. Employ CAPM to determine the cost of equity for DentHealth and the new investment and employ the relevant data in the question to determine the cost of debt and the WACC of the company and the new investment project.
3. CAPM only considers systematic risk. Explain this concept of risk and indicate why only systematic risk is relevant. Indicate whether you consider this concept of risk to be relevant in this question.
4. Assume that the beta of a similar company to DentHealth had an R^2 of 60%. Indicate what this means. Also, calculate the total beta for the company.

5. What other possible adjustments should Janet perhaps consider in determining the cost of equity of DentHealth?
6. Explain reasons why HPE, as a private equity company, may have a required return of 20%.

Page 7-58 **Question 7.32**

The BV Group is an international services, trading and distribution company. The group is one of the leading freight management companies in South Africa and operates terminals and focuses on logistics, freight and clearing services. The group is considering the acquisition of a similar but smaller independent logistics company, Intermove (Pty) Ltd. The target company's most recent Statement of Comprehensive Income and Statement of Financial Position are set out in Exhibits 1 and 2.

Exhibit 1

Statement of Comprehensive Income	20x1
for the year ended 30 June	Rm
Revenue	1,126.0
Operating expenses (excl. depreciation)	-1,059.3
EBITDA	66.7
Depreciation	-12.5
EBIT	54.2
Net interest payable	-10.3
Profit before tax	43.9
Tax	-10.4
Profit after tax	33.5
Retained earnings at beginning of year	109.2
Dividends	-11.8
Retained earnings at end of year	130.9

Intermove has 33.6 million shares in issue of which there are 3.2 million treasury shares owned by a subsidiary. The yield on the 10-year RSA government bond is 7.9%. The market risk premium is 5.5% and the company's beta is estimated to be 1.1. The capital structure of the company is represented by a debt-to-equity ratio of 44%, which reflects the current balance sheet book values of Intermove's debt (including short-term deposits) and equity.

The cost of debt is based on the current yield to maturity (YTM) of the bonds issued by ABC Ltd which is a comparable company. ABC's major bond issue is currently

trading at R106.34 per bond. The face value of each bond is R100 and the coupon rate is 12% per year, payable annually in arrears on 30 June. The redemption date is 30 June 20x5. The current date is 1 July 20x1 and the company's financial year-end is 30 June.

Sales revenue is expected to grow by 10% per year in 20x2 and 20x3 and then fall to a sustainable growth rate of 6% per year from 20x4. Operating expenses (before depreciation) is currently at 94% of sales but is expected to be 91.5% of sales in the future if the lease charges are excluded from operating expenses. The low margins reflect the competitive and service nature of the sector.

Lease charges for the year ending 30 June 20x1 amounted to R29m. The lease charge relates to plant and equipment and the lease charge is equal to the lease payment as this is a fixed annuity payment per year. The lease is therefore not subject to any escalation clause. The lease has a remaining term of 4 years. The company follows a policy of both owning and leasing assets and will continue with this policy in the future.

Depreciation in future years is expected to be 20% of the sum of opening book values of property, plant and equipment and intangibles. Goodwill is not expected to increase in future years. Investments constitute strategic non-operating assets and are expected not to change in the future. The expected net book value of property, plant and equipment, and intangibles in each year in relation to sales is expected to be as follows:

- Property, plant and equipment 9%
- Intangibles 1%

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Exhibit 2

Statement of Financial Position	20x1
as at 30 June	Rm
Non-current assets	
Property, Plant and Equipment	104.1
Intangibles	11.1
Goodwill	39.7
Investments (at fair value)	18.4
	173.3
Current assets	
Inventories	78.9

Receivables	107.0
Cash balances (operating)	11.3
Money-market deposits	22.5
	219.7
Total Assets	**393.0**
Equity	
Share capital	12.1
Retained earnings	130.9
	143.0
Non-current liabilities	
Borrowings	72.1
Post-retirement obligations	4.6
	76.7
Current Liabilities	
Short term loans and overdrafts	13.4
Taxation	11.0
Accounts payable & provisions	148.9
	173.3
Total Equity and Liabilities	**393.0**

Post-retirement obligations relate to the market value of post-retirement medical obligations less a defined benefit pension surplus. This represents the net underfunded position. Current contributions to the fund will not change in the future. In terms of working capital, the company will maintain current levels of inventories, and receivables relative to sales. Operating cash balances are expected to be 1% of sales revenue. Taxation owing is expected to be 1.1% of sales and accounts payable and provisions together will amount to 12.4 % of sales revenue in the future.

The PV of lease payments is expected to amount to 8.5% of sales in the future and amortisation of the PV of leases (“right of use”) is expected to be 20% of the opening value of such right of use balances. You can assume that the tax deduction (excluding the financing component) is expected to be equal to the amortisation of the PV of lease payments (right of use) balances. Long-term borrowings carry a fixed interest rate of 11% per year which is payable in arrears on 30 June of each year. Short-term borrowings pay a variable rate of interest and short-term deposits earn interest at a variable rate. Up to 20x1, lease charges were included as an operating expense and the capitalisation of leases was not yet mandatory.

Required:

1. Determine the present value of the lease payments as at 30 June 20x1 and calculate the debtequity ratio of Intermove after the capitalisation of leases as at 30 June 20x1. Determine the WACC of Intermove before and after the capitalisation of leases. (Assume the current beta does not recognise the implicit leverage of leases.) Page 7-60
2. Value the ordinary shares in Intermove at 1 July 20x1. Set out the operating Free Cash Flows of the company over the next 4 years and use the Free Cash Flow (DCF) to the Firm Model to value the ordinary shares of the company.

Question 7.33

Kidzo (Pty) Ltd ('Kidzo') is a South African retailer of babies' and children's clothing. The company was founded in 1993 by Ms Jessica James CA(SA). In order to avoid liquidation arising from competitive pressures and supply chain disruptions, the company is considering diversifying its operations by manufacturing hand sanitisers.

According to the board minutes, the opportunity to invest in the manufacturing of hand sanitisers would be funded by a combination of existing cash, a lease entered into for the use of the bottling plant, and debt finance in the form of a bank overdraft to cover short-term fluctuations. No equity would be raised. Jessica proposed that the WACC therefore be determined on the basis of these three forms of finance.

Jessica noted that according to the bank statement, the company had R500 000 cash available to fund a portion of this project. In her opinion this amount would be free finance and the only loss would be the interest Kidzo would not be able to earn on these funds. The lease would be classified as an operating lease from the lessor's perspective, as the lessor would retain ownership of the asset. The directors Ronel Snyders and Thabiso Myoli, have determined that the right- of-use asset and corresponding liability value would be capitalised at a value of R2 million. The contractual rental payments for the lease would be R500 000 per annum, payable in arrears. The lease would have a term of five years. Jessica was unsure how the lease should be included in her WACC calculation. The company's long-term pre-tax cost of debt is on average 10% per annum (with overdraft financing incurring interest at 15% per annum).

Kidzo currently has a debt to total firm value ratio of 75%, although its optimal capital structure is a debt to total firm value ratio of 50%. The current levered beta coefficient of the retail sector on the JSE has been estimated to be 0.9, while the

levered beta of the chemicals industry on the JSE is 1.1. Kidzo has similar gearing levels to listed peers in both the retail and chemicals sectors.

Other information:

- Yield on South African government bonds
  - One-year 9.38%
  - Five-year 7.86%
- The historic market return over the past 25 years on the JSE has been 12.5%.
- Kidzo's effective corporate rate of taxation is 27%, which is not expected to change in the foreseeable future.
- Tax is paid in the financial year in which cash flows occur.

Required:

Calculate the estimated WACC to be used for the evaluation of the new hand sanitiser investment. Motivate the choices you make and provide reasons for the items you omit from your answer. [*SAICA ITC 2021 (adapted – this is limited to the section which requires the calculation of WACC)*]

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[1] In market parlance, these are called perpetual fixed-rate notes. However, almost all notes are issued at floating rates or offer a fixed rate for a number of years, followed by a floating rate.

[2] This is in line with section 24J of the Income Tax Act.

[3] In 2017, Edcon was restructured with debtholders swapping debt for equity due to the inability of Edcon to meet its loan obligations. The costs of financial distress for Edcon were real and the group was placed under business rescue in 2020. Another case of financial distress relates to Lonmin which was acquired by Sibanye due to its poor capital structure.

[4] If the debt–equity ratio is X, then the debt ratio is [X/(1 + X)], so that the debt ratio is 1.22/(1 + 1.22) = 0.55.

[5] If the current beta already includes the operating leases in the debt-to-equity ratio, then WACC will fall to 10.37% [0.14 x 0.4587 + 0.073 x 0.5413]. All this means is that the current WACC is overstated at 11.77%, because it overstates the equity component of the firm's capital structure.

[6] Investing in Russian bonds would not have been a good idea either.

[7] Damodaran at NYU produces equity risk premiums for countries by adjusting the implied market risk premium of the USA S&P500 by a country risk premium. In January 2023, this resulted in a market risk premium of 10.44% or 11.13%. However, the country risk premium is implicit in the RSA government bond yields and therefore it is relevant not to double count for country risk. A quick approximation using an RSA bond yield of 10.5% plus an ERP of 5.1%

offers a cost of equity of 15.6%. Applying a USA government bond yield of 3.8% at the time of writing plus 10.8% (average of 10.44% and 11.13%) results in a cost of equity of 14.6% which is not far from our initial estimate. Do not however add the Damodaran estimated market risk premium for South Africa to the RSA government bond yield.

[8] The 2022 PwC Survey found that 76% of respondents adjusted the CAPM cost of equity by a small company risk premium which varies based on the size of the company. The small size premium varies from an average of 5.2% (0 - R250m), 4.4% (R251m-R500m), 3.4% (R501m-R1000m) and 2.4% (R1000m-R1500m). Size refers to equity market value. Matthew Human, a partner at PwC who specialises in business valuations, writes that he does not take a strict view around equity values as an indicator of size and the related small company risks. Matthew considers the small company risks of a business in terms of the extent to which small company risks are present in that business's cash flows. He states that in practice he "considers enterprise values, equity values and also other factors, such as access to financial markets, market position, and geographical diversification in forming a view around what small share premium he believes is appropriate in the specific circumstance".

[9] Aswath Damodaran, (2006), Damodaran on Valuation: Security Analysis for Investment and Corporate Finance, 2nd edition, Wiley Finance series, John Wiley & Sons, see pages 57–59 on Total Beta.

[10] Alternatively, we can present the beta as $\beta = \rho(\sigma_s/\sigma_m)$ where p is the correlation co-efficient and Total Beta = $\beta/\rho = (\sigma_s/\sigma_m)$.

[11] In an efficient market, the market value of equity will be determined by the market value of a company's assets less the market value of its liabilities.

[12] From January 1998 until December 2022, the MSCI Small Caps Index outperformed the MSCI World Index by 3.14% per year.

[13] Ken French maintains a website: http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/index.html in which he sets out the data for the USA and other countries, but unfortunately not for South Africa.

[14] However, there seems to be mean reversion after about 4 years.

[15] Okyere-Boakye, K & O'Malley, B. (2016). Downside CAPM: The Case of South Africa. *Journal of Economic and Financial Sciences* 9(2), 578–608.